

7. Current Solar System

2025 October 27

System: in astronomy, • a group of celestial objects connected by their mutual attractive forces, especially moving in orbits about a center. (New Oxford American Dictionary)

Today's Lecture – Introduction

- In the previous lecture, I explained the atmospheres of (1) gas planets and (2) terrestrial planets, and compared the atmospheric compositions of (2-1) Venus and Mars and (2-2) Earth.
- I ran out of time during that explanation, so today I will review and continue this topic of planetary atmosphere formation.
- Understanding how an atmosphere forms and what it is made of is essential for studying habitable planets, especially how a habitable region on the surface can exist.
- In the second half, I will mention the importance of tidal forces. It can generate internal heating and may create subsurface habitable regions in some satellites.
- By the end of today, you should be able to explain why atmosphere formation is critical for surface habitability, and how tides can enable habitability underground.

Formation of Atmospheres on the terrestrial planets (2)

Formation of Magma Planets

- During the formation of the Solar System, **protoplanets frequently collided** with other objects.
- The energy from **impacts** and **radioactive decay** heated the planets enough to melt them entirely, forming **magma oceans**.
- Water contained in the rocks **evaporated** and formed a dense atmosphere of water vapor.
- The early atmospheres of Earth were **hundreds of times denser** than today's atmosphere.



<https://earthsky.org/space/missing-impact-debris-planetesimals-solar-system/>

Formation of Atmospheres on the terrestrial planets (1)

Earth: From Magma to Ocean Planet

- The young Earth was **completely molten at first**.
- As it radiated heat into space, the surface **gradually cooled** and **solidified over a few million years**.
- As the surface cooled, **water vapor condensed** and fell as torrential (heavy) **rain**.
- This rain was so intense that **the oceans formed in a short period of time**.
- Repeated cycles of rain and evaporation cooled the planet further, leading to a **stable, ocean-covered Earth** with a more moderate atmosphere.

Formation of Atmospheres on the terrestrial planets (3)

Venus: The Runaway Greenhouse Planet

- For **Venus**, it took **~100 million years** for the magma surface to cool — much longer than Earth.
- During this time, **water vapor remained in the atmosphere** instead of condensing into rain.
- Strong **solar ultraviolet radiation decomposed H₂O** into **hydrogen and oxygen**.
 - Hydrogen escaped easily to space (too light).
 - Oxygen reacted with surface rocks and was absorbed into the crust.

Formation of Atmospheres (4)

Why the Terrestrial Planets Lost H and He

- The terrestrial planets are too small to retain H and He in their atmospheres.
- Although H and He are the most abundant elements in the Universe, they are almost absent from Earth's atmosphere.
- → Because H and He atoms are very light, they move fast and easily escape into space at typical atmospheric temperatures.

Rebuilding of Secondary Atmospheres

- After losing most of the original H and He, new atmospheres formed by volcanic outgassing and impacts.
- These secondary atmospheres mainly contained:
 - H₂O and CO₂ (major components)
 - N₂ (minor component)

Formation of Atmospheres (5)

Early Earth's Dense CO₂ Atmosphere

- The primordial Earth atmosphere likely had ~1000 times more CO₂ than today.
- Where did it go?
 - H₂O → condensed to form **the oceans**
 - CO₂ → **dissolved into the oceans** (탄산수)
 - and **precipitated as carbonates** (limestone, 석회암)

Formation of Atmospheres (6)

- By contrast, N_2 is chemically inert. It can remain in the atmosphere.
- The second major constituent of the present-day Earth's atmosphere is O_2 .
- It is primarily produced by photosynthesis (광합성) in plants.
- Ozone (O_3) forms in the stratosphere from O_2 interacting with solar UV photons. It blocks UV photons from reaching the ground.
- This makes it possible for life to survive on land by shielding it from harmful solar UV radiation.
- The presence of O_2 and O_3 in our atmosphere is considered a strong indicator of life.

Temperatures of inner objects in the solar system

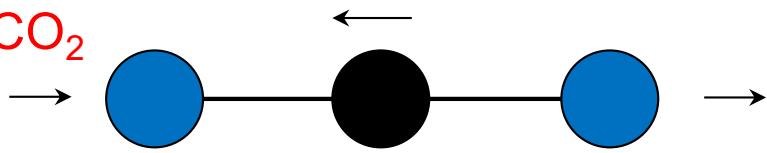
	Distance from the Sun (au)	Blackbody temperature (K, deg C)	Albedo*	Reflective cooling (K)	Greenhouse warming (K)	Estimated temperature (K, deg C)
Mercury	0.4	448 (175)	0.07	-8	0	440 (167)
Venus	0.7	328 (55)	0.90	-144	+510-550	737 (464)
Earth	1.0	278 (5)	0.30	-25	+33-35	288 (15)
Moon	1.0	278 (5)	0.11	-7	0	271 (-2)
Mars	1.5	226 (-47)	0.25	-16	+5	215 (-58)

*Fraction of reflected sunlight relative to the incident sunlight.

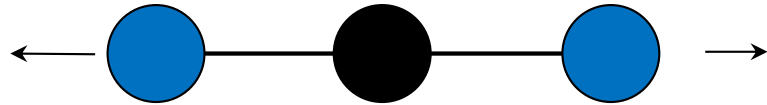
Greenhouse Effect

- The atmosphere is transparent to visible light but mostly opaque to infrared.
- The Infrared opacity comes from absorption bands of H_2O , CO_2 , CH_4 , and other molecules. Photons are absorbed by the ground, heating it up. The warm ground radiates electromagnetic energy mainly in the infrared wavelengths..
- **The Greenhouse Effect** is responsible for making the Earth about 35 K warmer than it would be if it had no atmosphere.
- Therefore, without the Greenhouse Effect, there would be no liquid water on the Earth, only ice. Since life as we know it requires liquid water, the Earth would be inhospitable to life without the Greenhouse Effect.

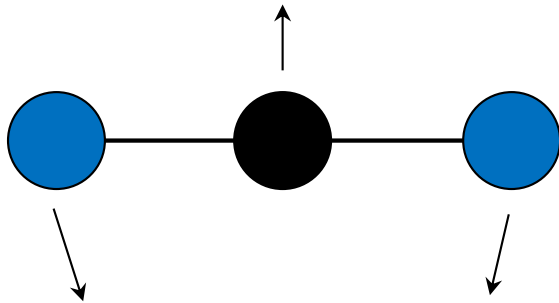
CO₂



asymmetric stretching 4.35 μm

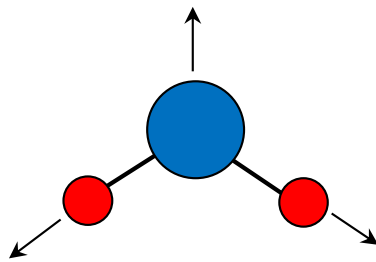


symmetric stretching 7.46 μm

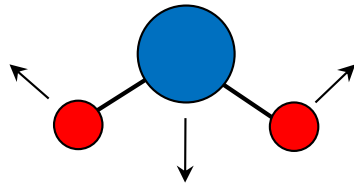


bending 15.02 μm

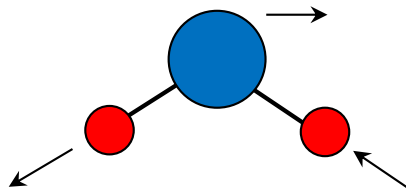
H₂O



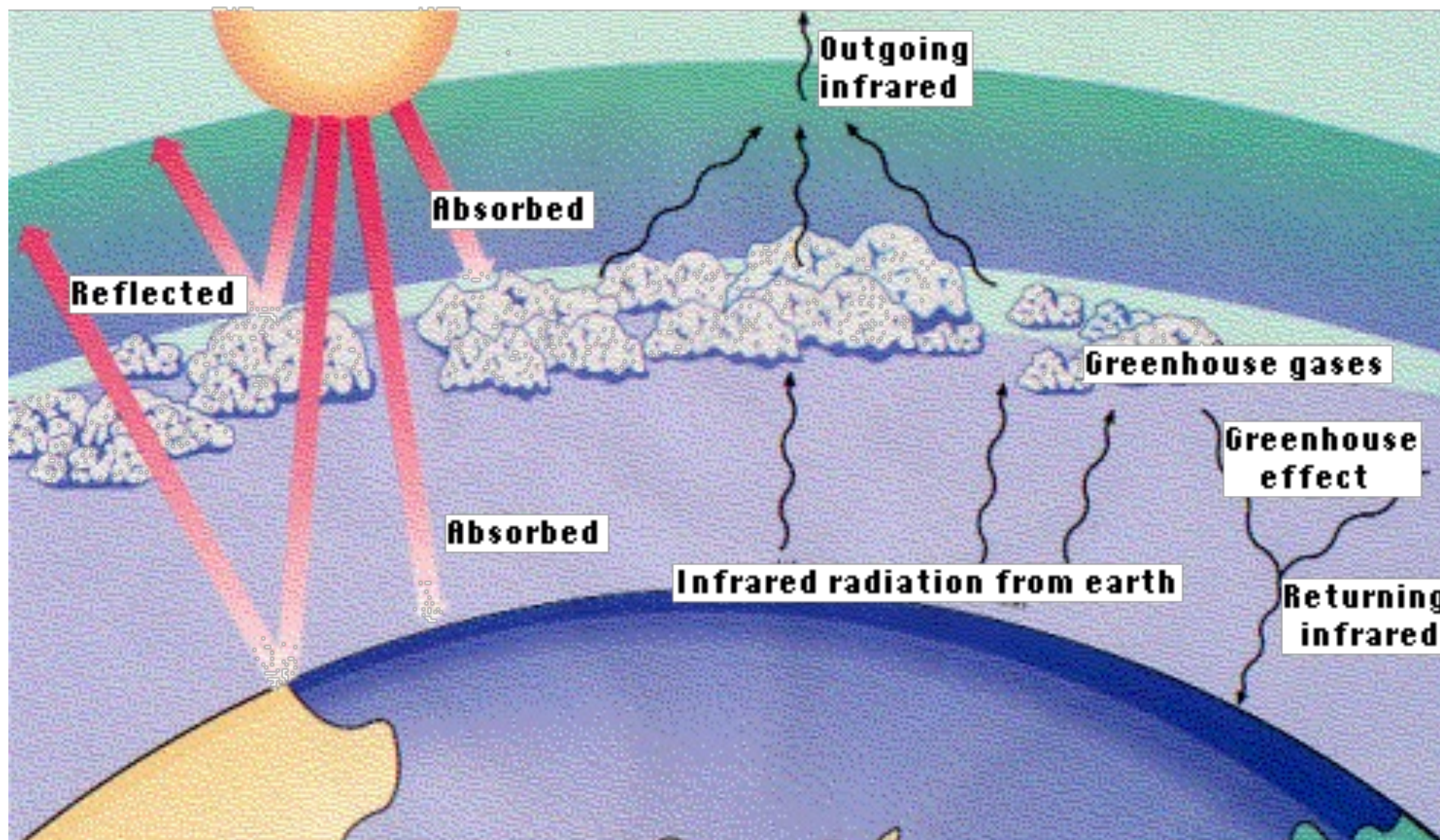
symmetric stretching 2.74 μm



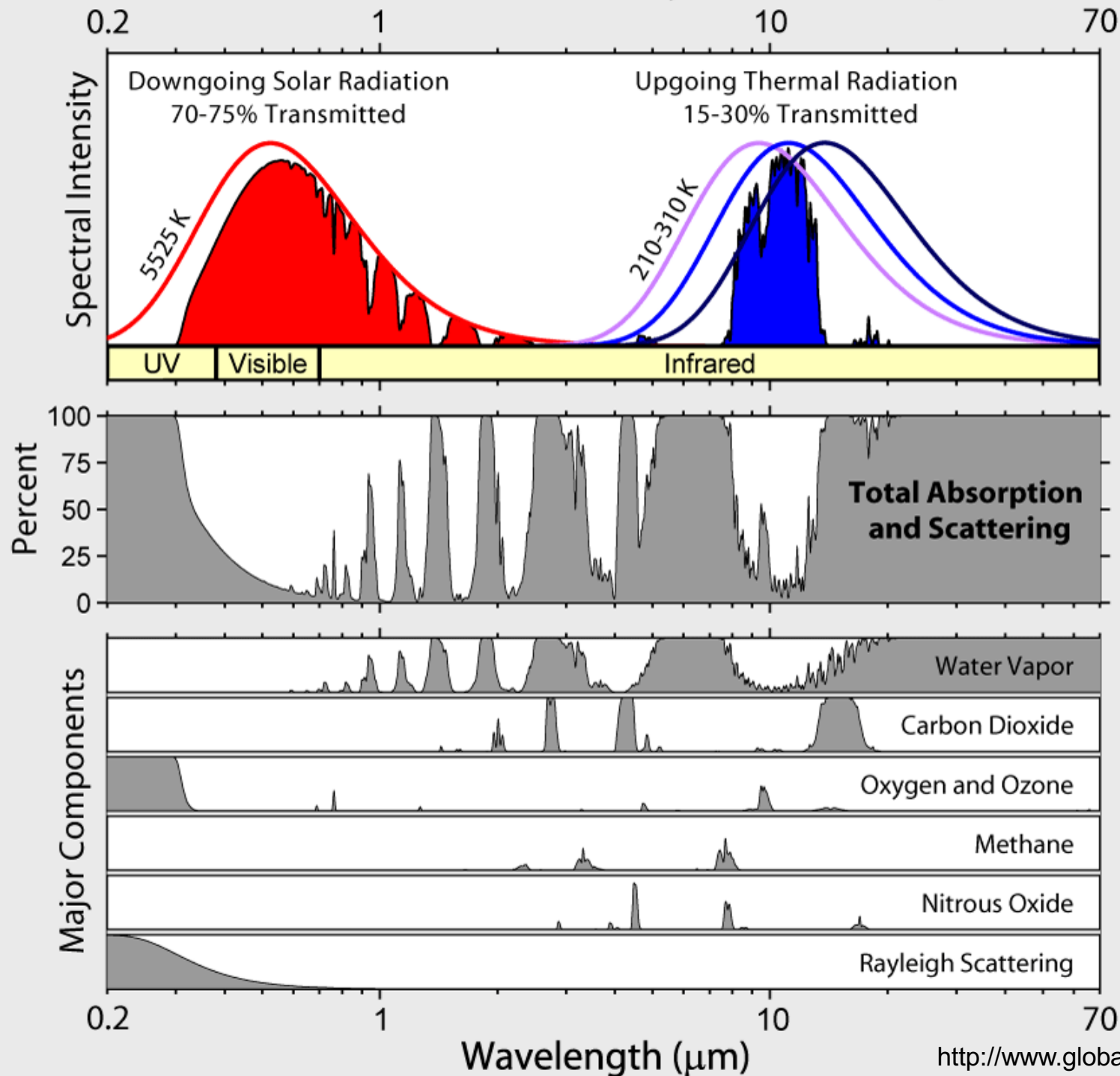
symmetric bending 6.27 μm



asymmetric stretch 2.66 μm



Radiation Transmitted by the Atmosphere

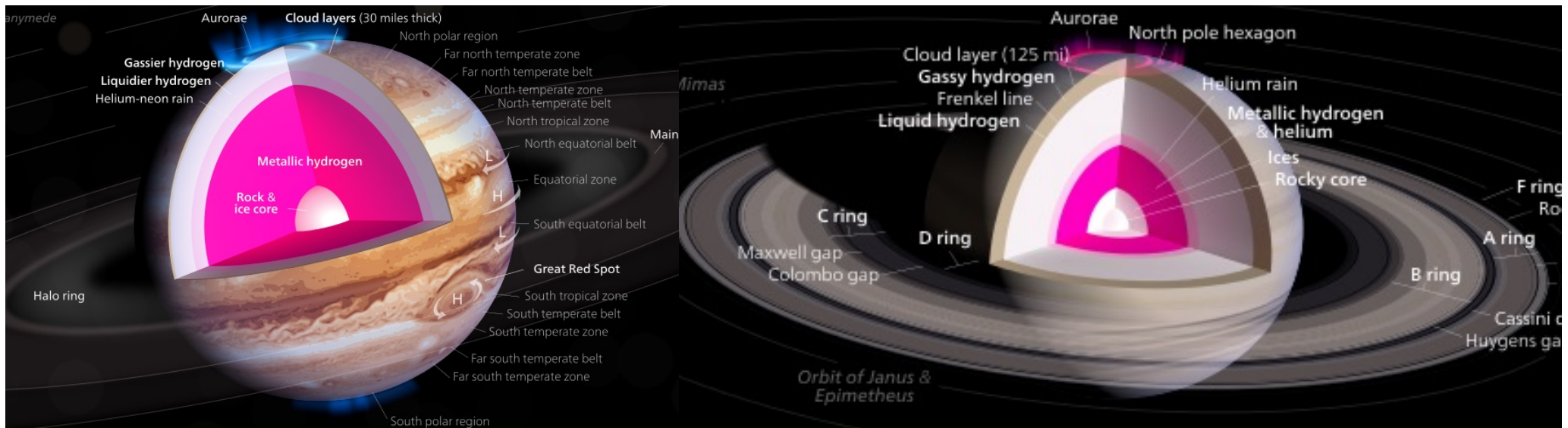


Wien's
displacement
law

$T \times \lambda = 3000$
 $\uparrow$ $\uparrow$
 K μm

Gas giant planets: Jupiter & Saturn

- **Dominant composition:** Hydrogen + Helium, like the sun
- Surface clouds: ammonia ice, water ice....
- Even deeper: rocky core of $\sim 10\sim15 \times$ (Earth mass)



Jupiter and Jovian System

- 5th planet from the Sun, the largest planet in the solar system.
- >60 known satellites.
- The planet is composed of mostly hydrogen and helium just like the Sun.
- Colorful bands parallel to the Equator. Lot's of swirls and spots: storms. One storm, the "Great Red Spot" is 2 times the size of Earth, and has lasted for at least 300 years. It's a gigantic hurricane.
- The planet appears oblate (squashed). This is due to rapid rotation (once every 10 hours!).

Jupiter's Moons in Motion
From left to right: Europa, Io, Callisto
North up

13/05/2008 16:21 - 17:55 UT
Central Coast, NSW Australia
12" newt + 5x powermate + DMK21AU04

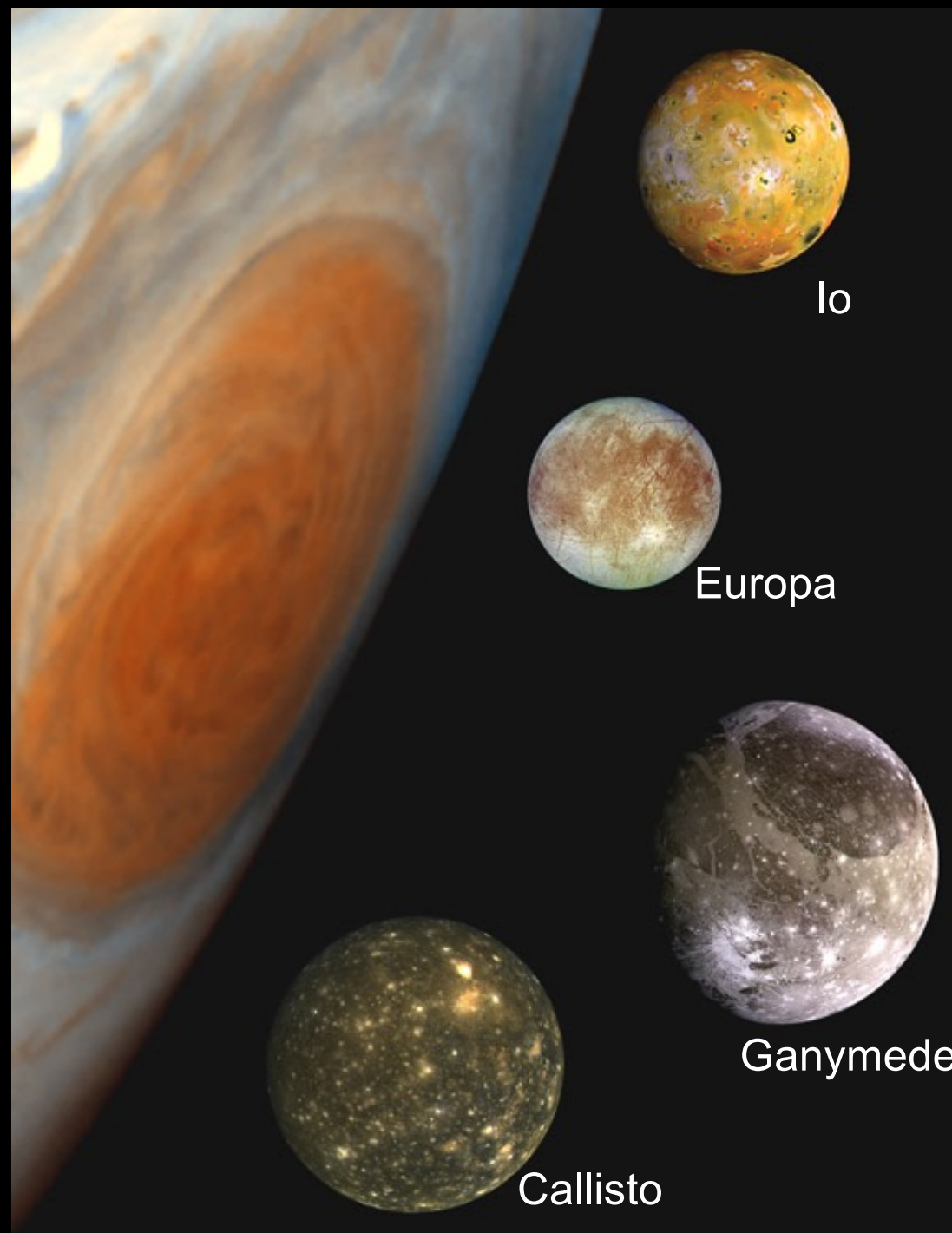
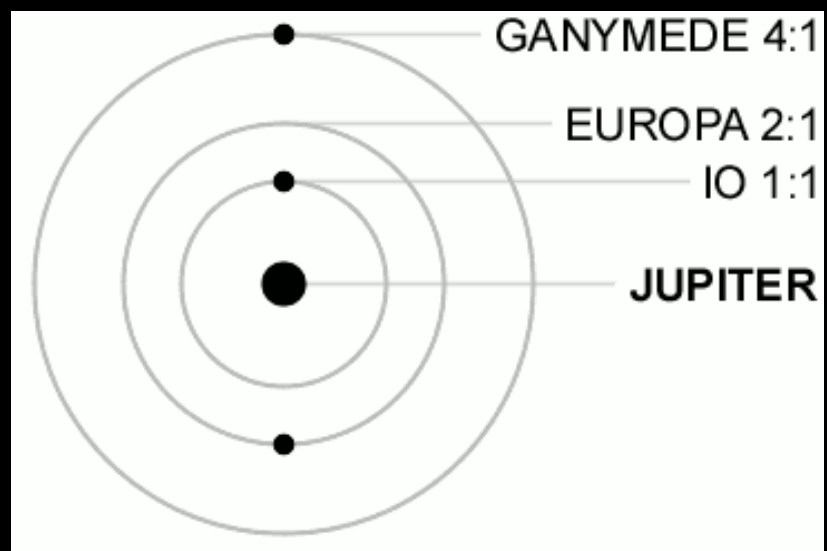
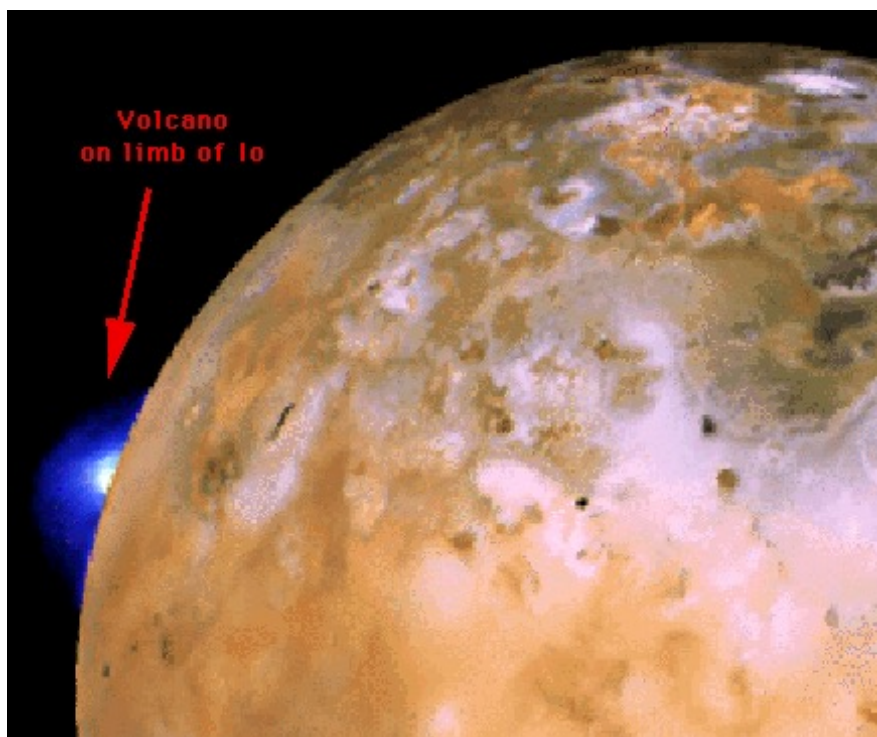
Mike Salway
www.iceinspace.com.au



HST/NASA

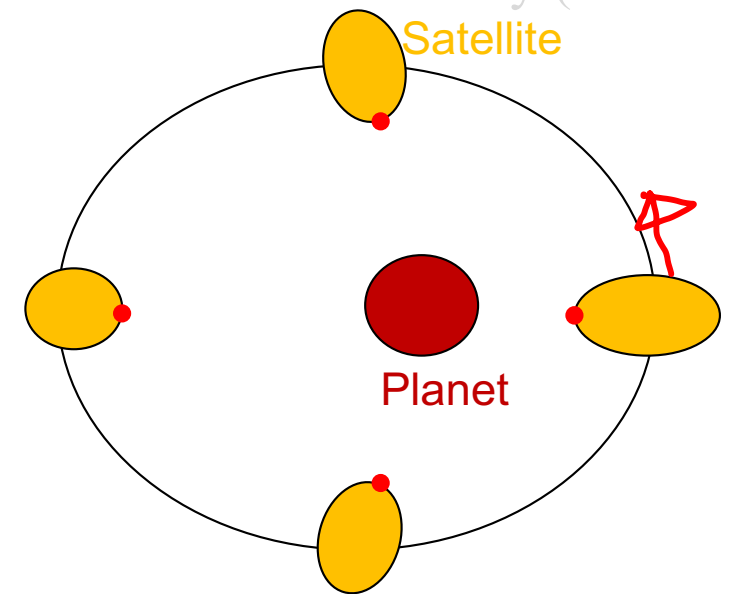


- Jupiter has very thin rings
- The Galilean Satellites: the four largest satellites of Jupiter
 - **Ganymede**: The largest moon in the Solar System. It has an old, cratered surface.
 - **Callisto**: Heavily cratered like Ganymede. Icy surface
 - **Europa**: Smooth surface with narrow, dark stripes running across, few craters. The surface seems to be fractured ice. Recent observations by the Galileo spacecraft suggest that there would be a worldwide ocean of liquid water beneath the frozen surface. Life?
 - **Io**: **Erupting >100 volcanoes!** The interior is molten; this is due to tidal forces.
- The other known moons are small; many are captured asteroids or comets.



Tidal Force

- The tidal force is a secondary effect of the gravitational force. It arises because the gravitational force exerted by one body on another is not constant across its diameter.
- Usually, one side of a satellite faces a planet. For example, when you observe the Moon, you always see only the same side. Why?
- This is because the shape of the satellite is slightly distorted from a spherical shape due to **tidal forces**. This distortion causes the satellite's rotation period to coincide with the orbital period. As a result, only one side can be seen (**tidal locking**).
- Galilean satellites are also tidally locked to Jupiter. The orbital eccentricity (non-zero) causes the time-variation of tidal force.
- In the case of Io, the vertical bulge of the surface varies by 100 m between the nearest and farthest points to Jupiter. Because the orbital period is only 43 hours, Io's shape has kept frequently changing as time goes on!





The Moon's appearance changes throughout the month.
Credit: NASA's Scientific Visualization Studio

Tidal bulge



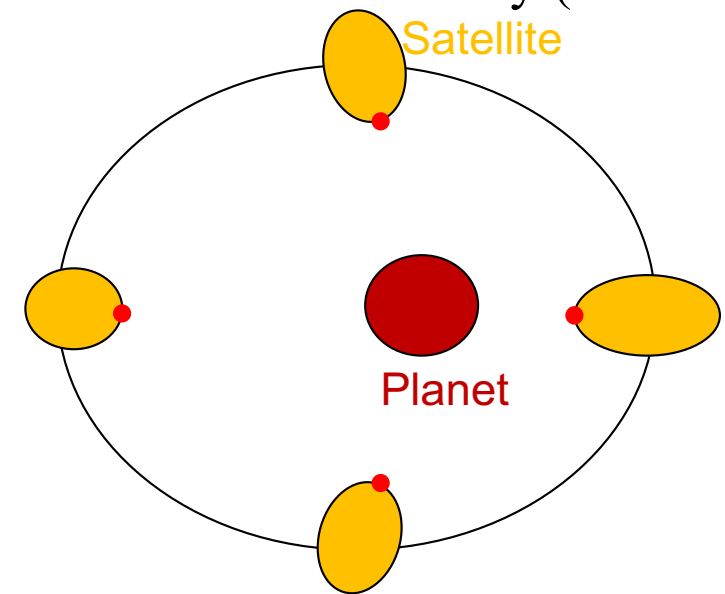
→
Moon's gravity

Tidal bulge due to inertia

Tidal bulge due to gravity

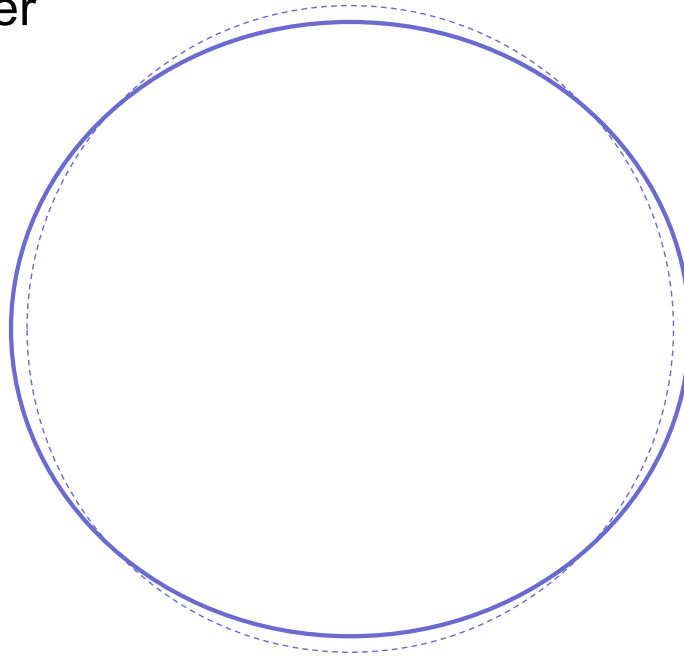
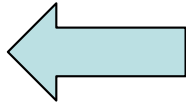
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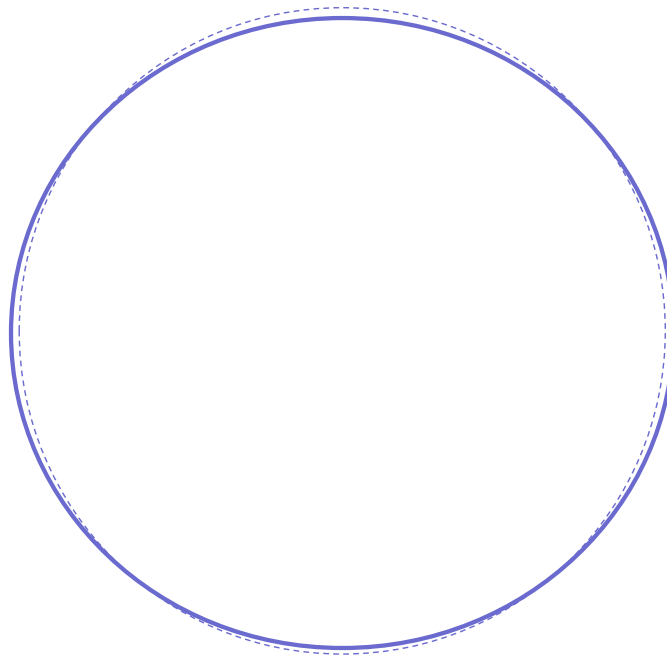
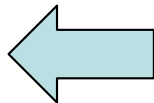


Tidal deformation

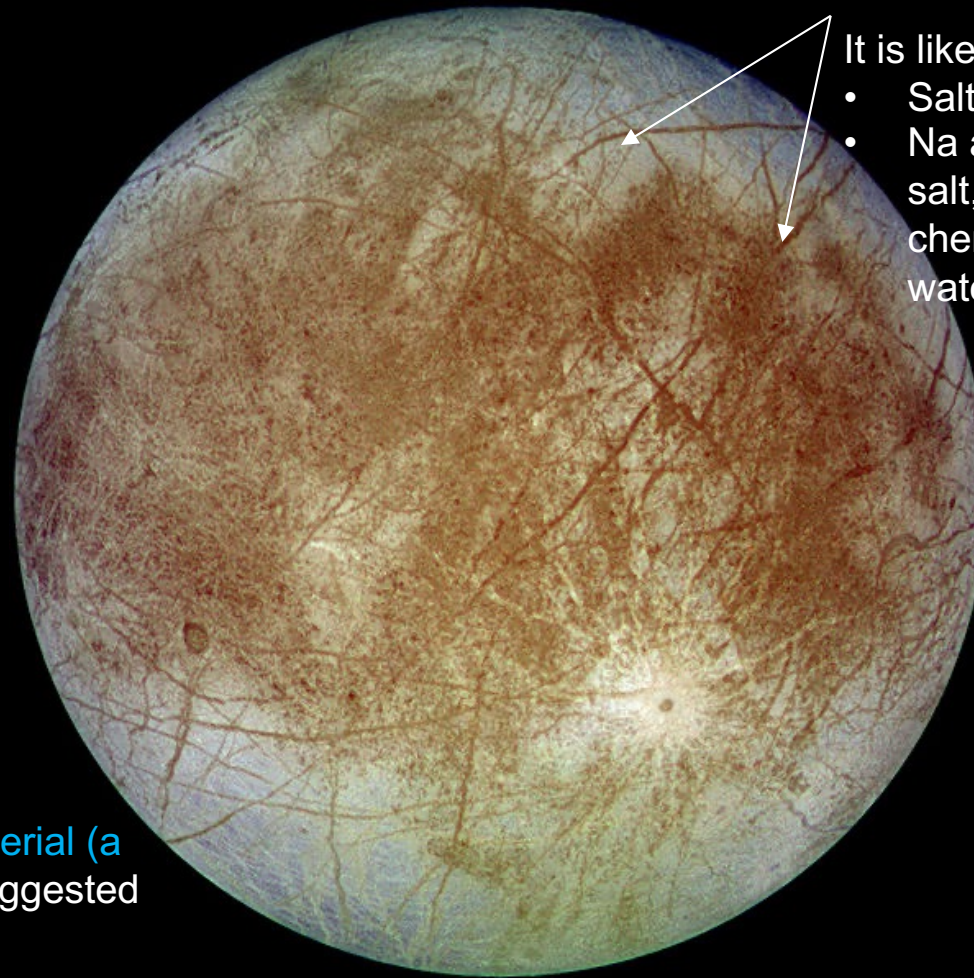
When it is close from Jupiter



When it is far from Jupiter



Europa (Moon of Jupiter)



Na and Mg

It is likely that:

- Salty water spurt out
- Na and Mg, the source of salt, can be formed by chemical reactions between water and rock.

The existence of **conductive material (a high electrical conductivity)** is suggested by magnetic field observation.

- Rocks and ice cannot explain the conductivity.
- Seawater (with abundant ions) relatively close to the surface would be a possible interpretation.

Surface temperature

50-125 K

Surface pressure

0.1 micro Pa

Size

0.245 Earth radius

Jovian Satellite: Europa



- Scientists consider that there would be liquid water beneath Europa's surface.
- The surface temperature is ~ 110 K at the equator, keeping Europa's icy crust.
- This heat energy is associated with tidal flexing.
- Some scientists interpret 'chaos terrain' as a region where the subsurface ocean has melted through the icy crust.

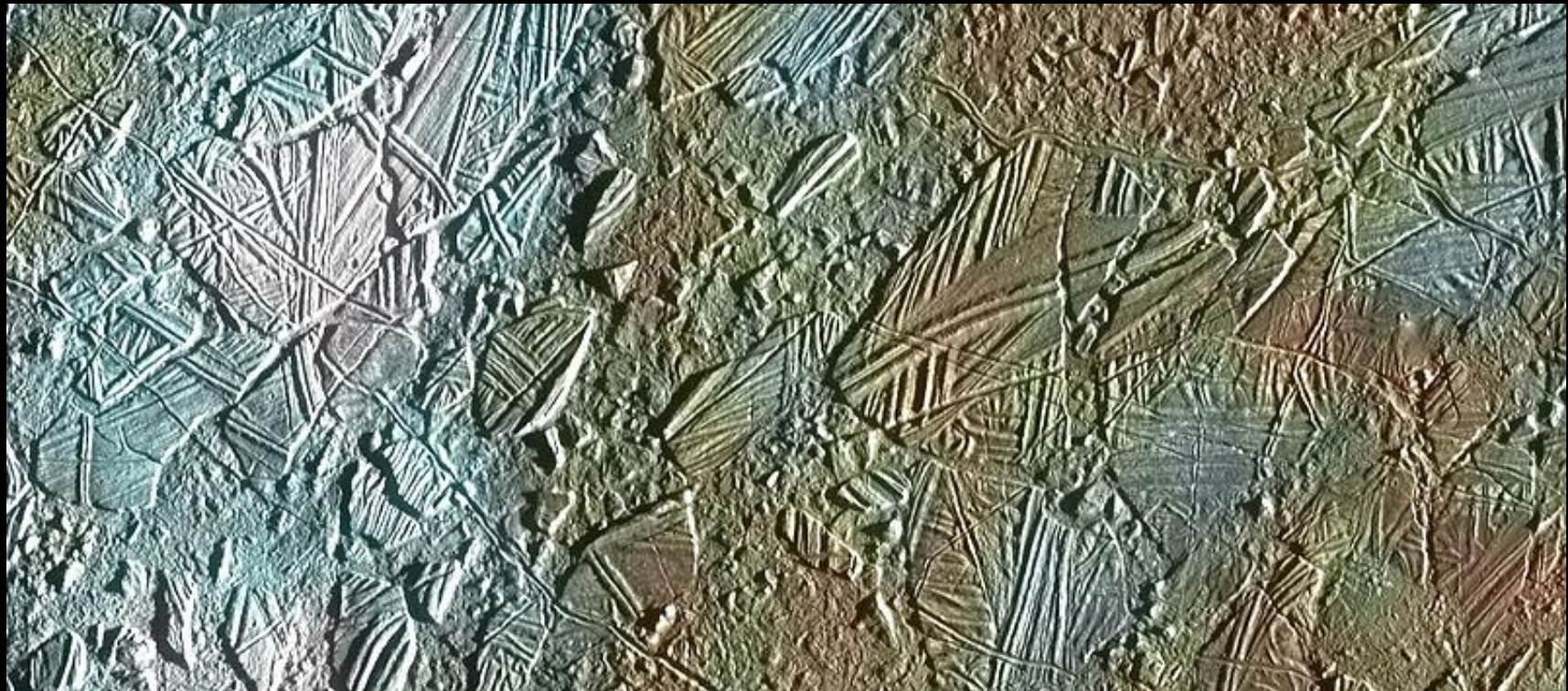
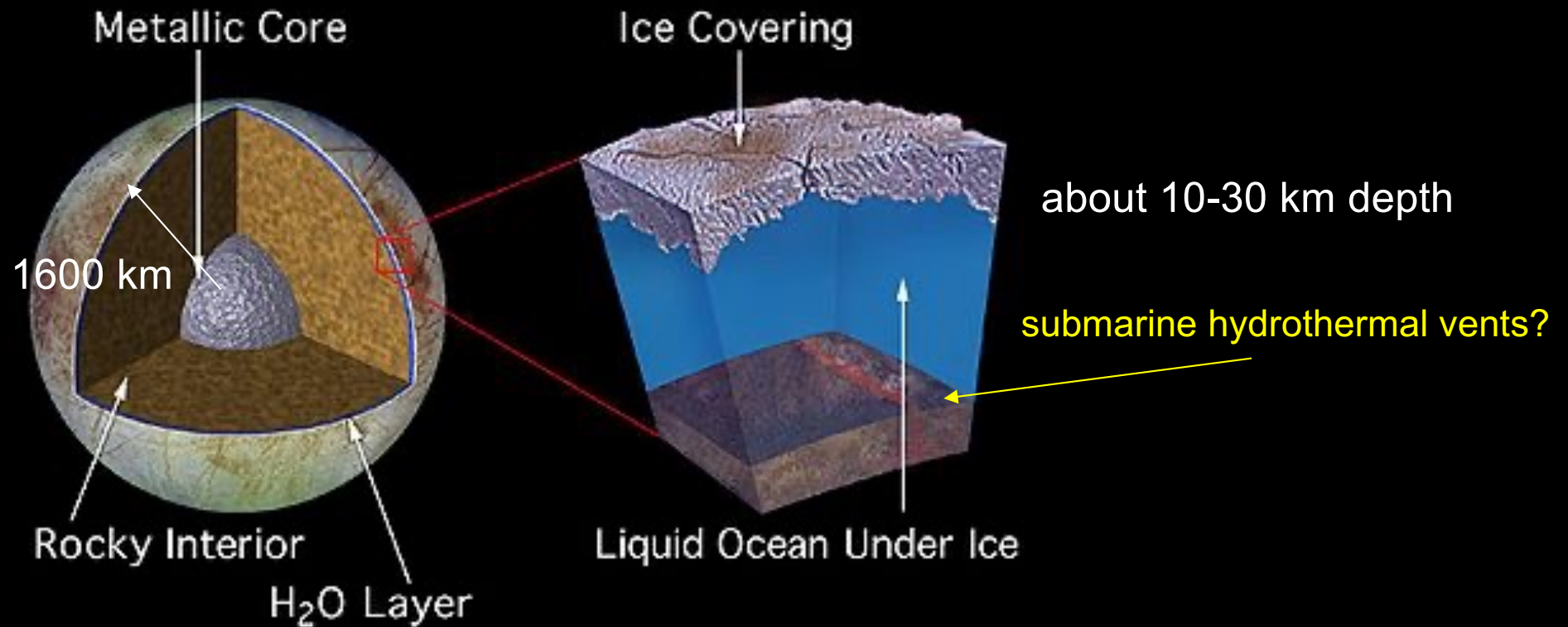


Image credit: NASA

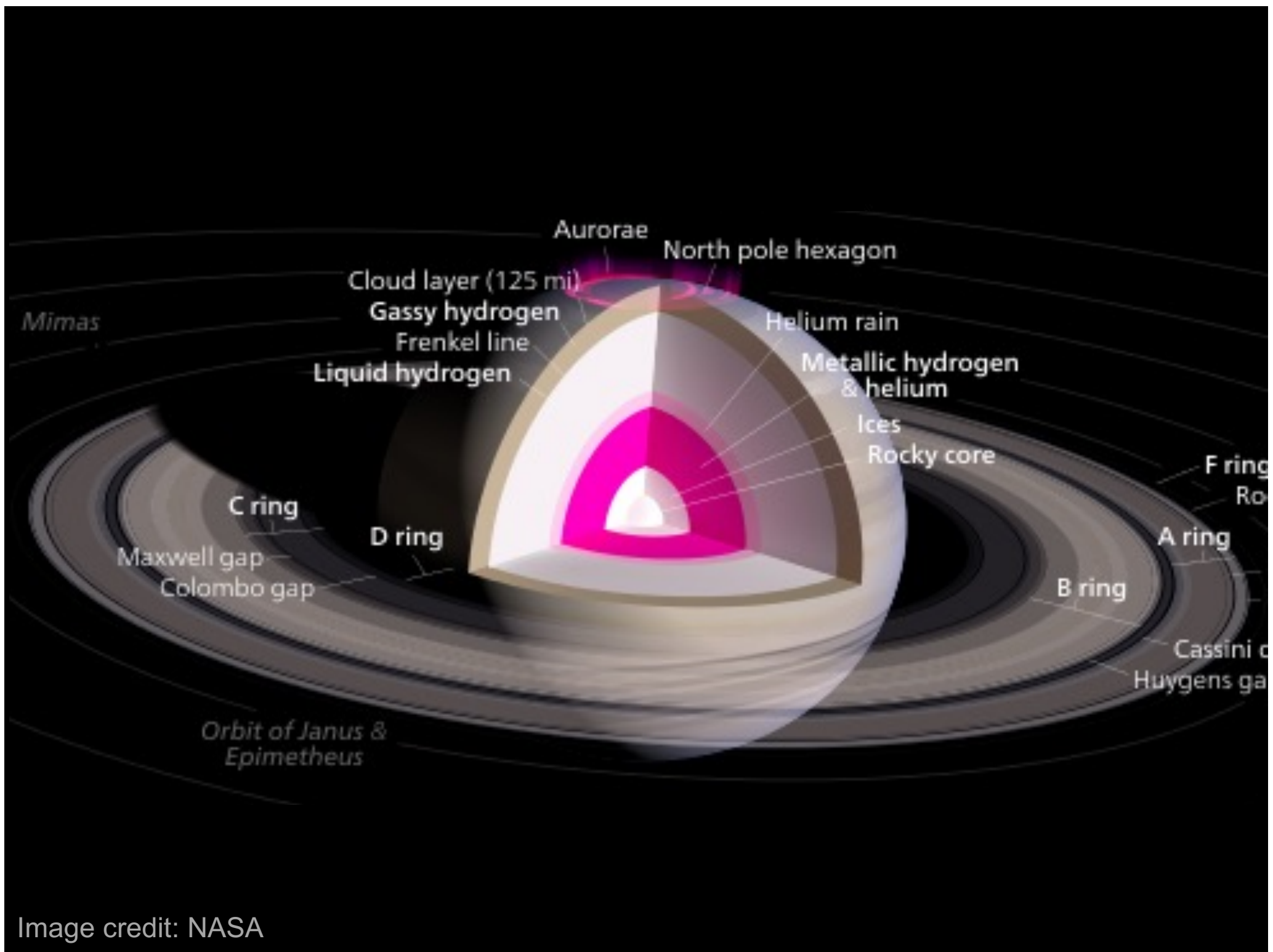
Mass density of Europe: 3.0 g/cm^3



A proposed model inside Europe

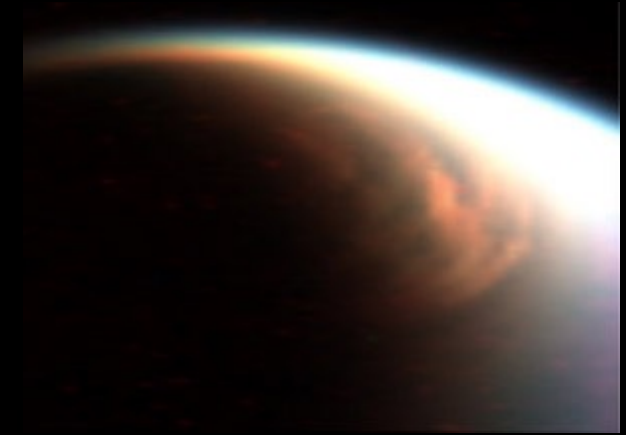
Saturn and Saturnian System

- 6th planet. Saturn has a remarkably low density ($<1 \text{ g/cm}^3$).
- Like Jupiter and the Sun, it is mostly composed of Hydrogen and Helium, with an Earth-like core.
- Its most outstanding feature is its magnificent rings. The rings are made of chunks of rock and ice. A catastrophic impact likely fractured a 500 km satellite to produce the ring.
- Saturn has more than 146 satellites. The largest satellite, Titan, has a thick Nitrogen atmosphere like Earth. There are methane lakes and rain of organic materials on the surface.



Titan (Moon of Saturn)

The 2nd largest satellite in the solar system



False color image of the cloud on the pole



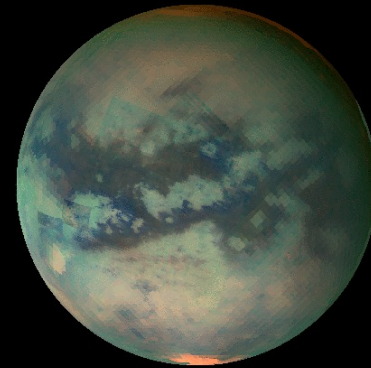
The only know celestial object with the definitive evidence for liquid on its surface

Surface temperature	94 K
Surface pressure	147 kPa (1.4 atm)
Atmospheric composition	N ₂ 98.4 %
	CH ₄ 1.4 %
Diameter	5150 km

13,000 km



5,000 km



(Infrared image)

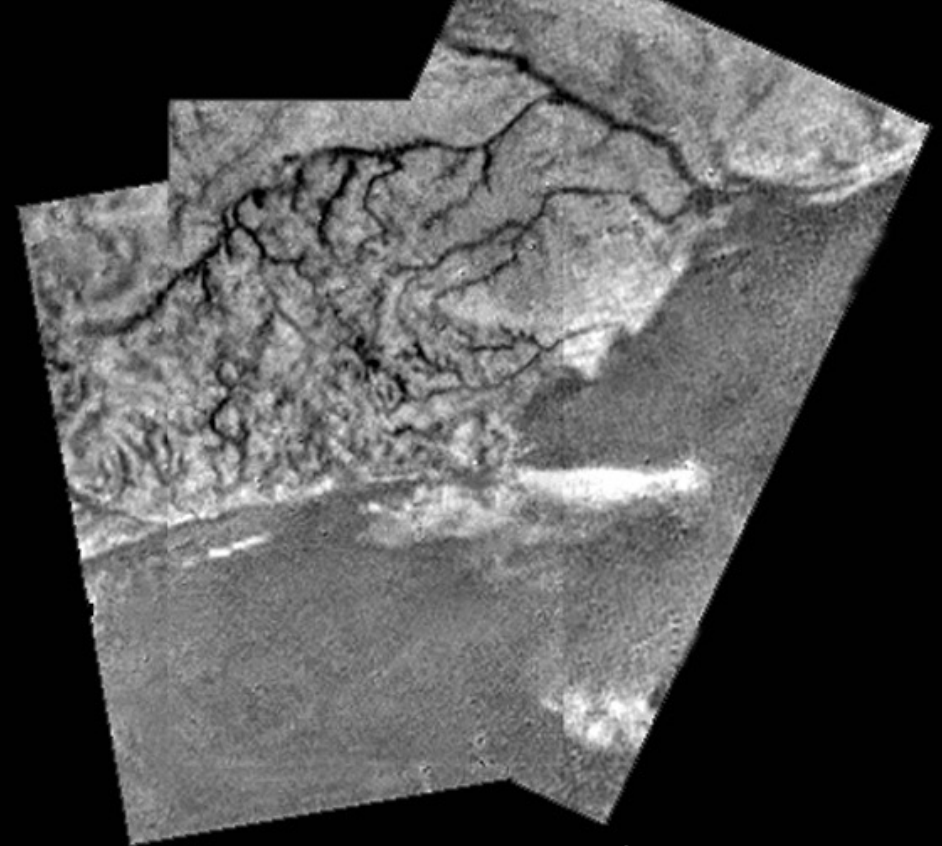
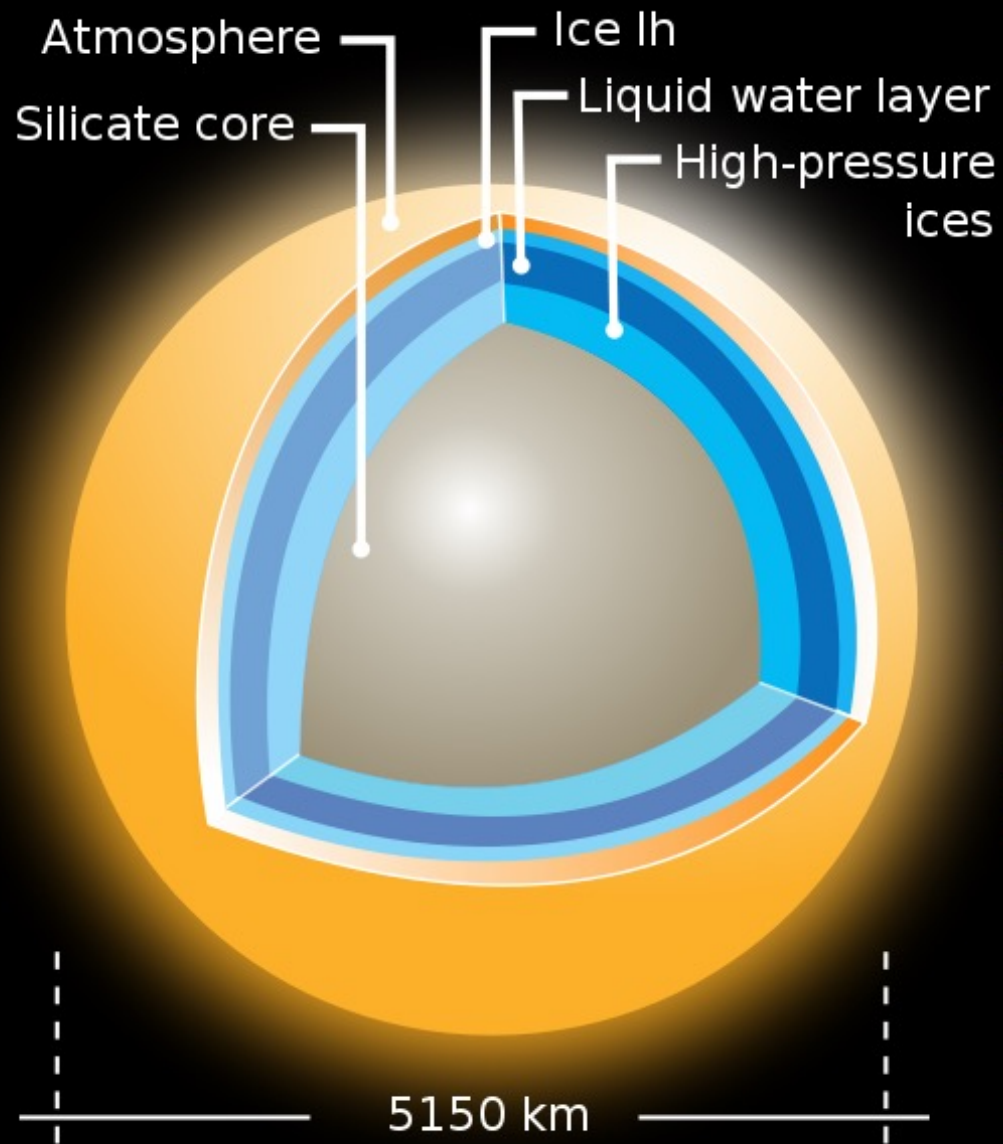
3,500 km



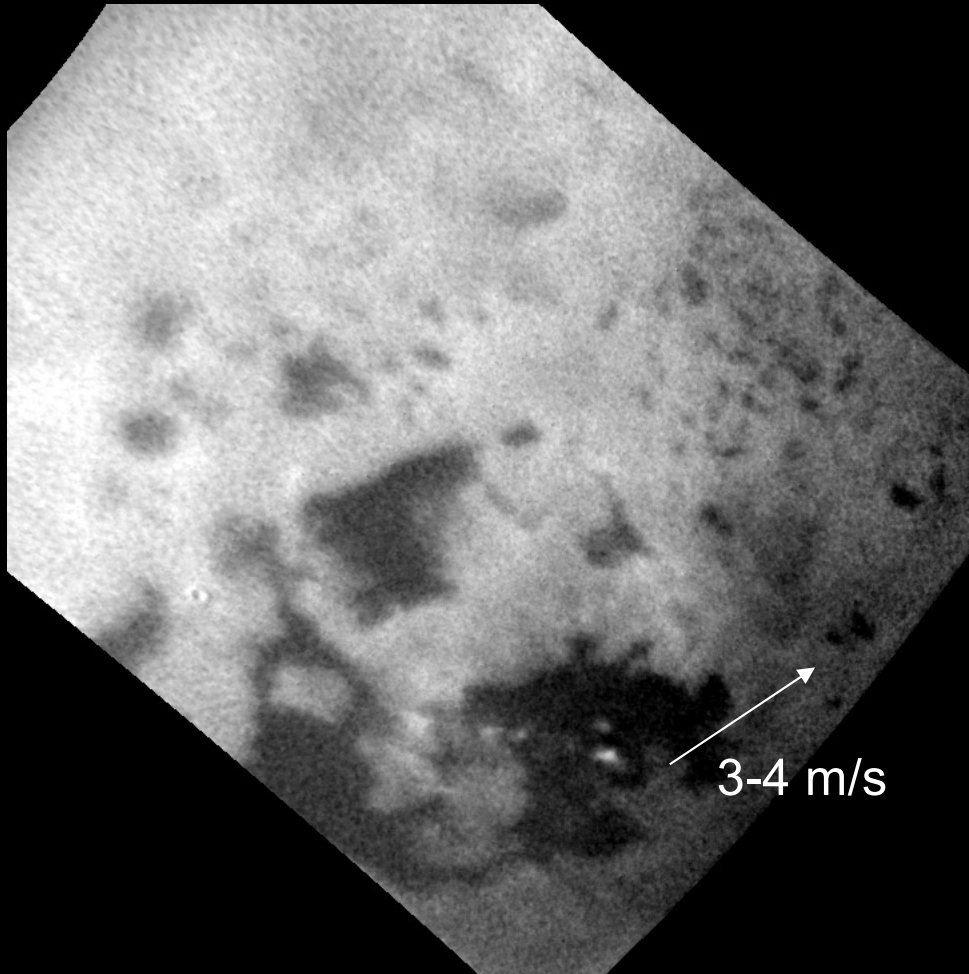
Titan is

- the largest moon of Saturn
- the only natural satellite known to have a **dense atmosphere**
- and the only object other than Earth (and Mars) for **clear evidence of surface liquid**

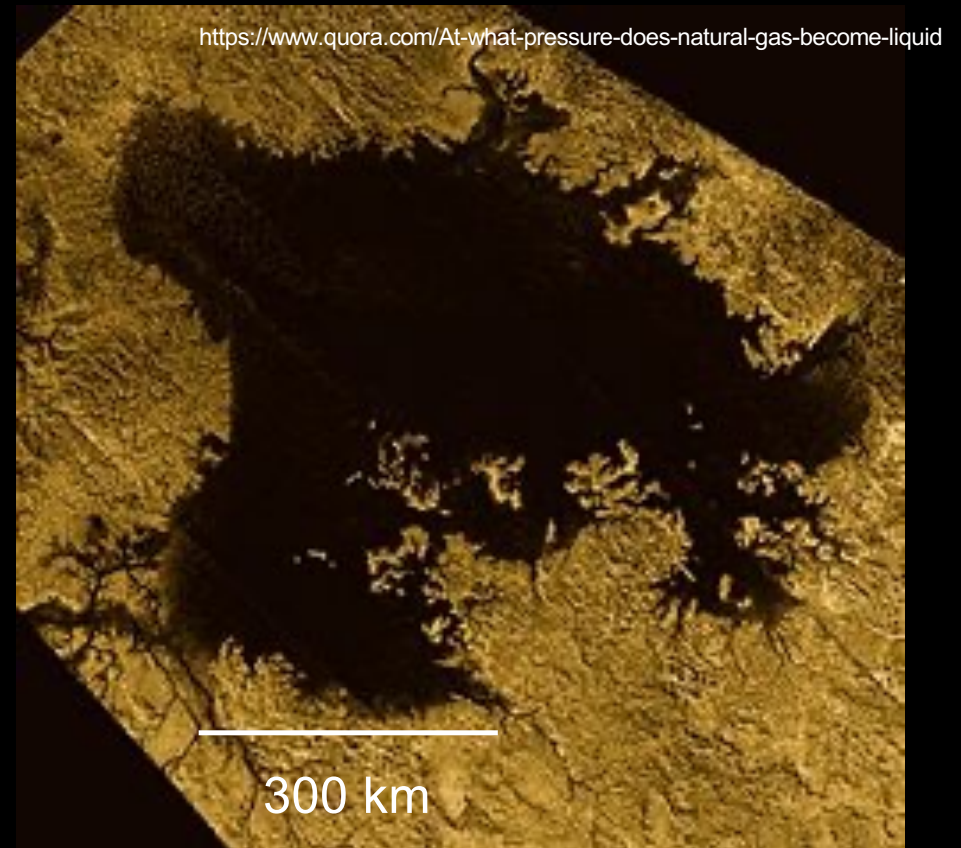
Image credit: NASA/ESA



The Earth has a variety of landforms and meteorological phenomena formed by the water cycle (the atmosphere, clouds, oceans, rivers, and lakes). Under the extremely low temperature of <100 K, Titan has a similar cycle due to methane.

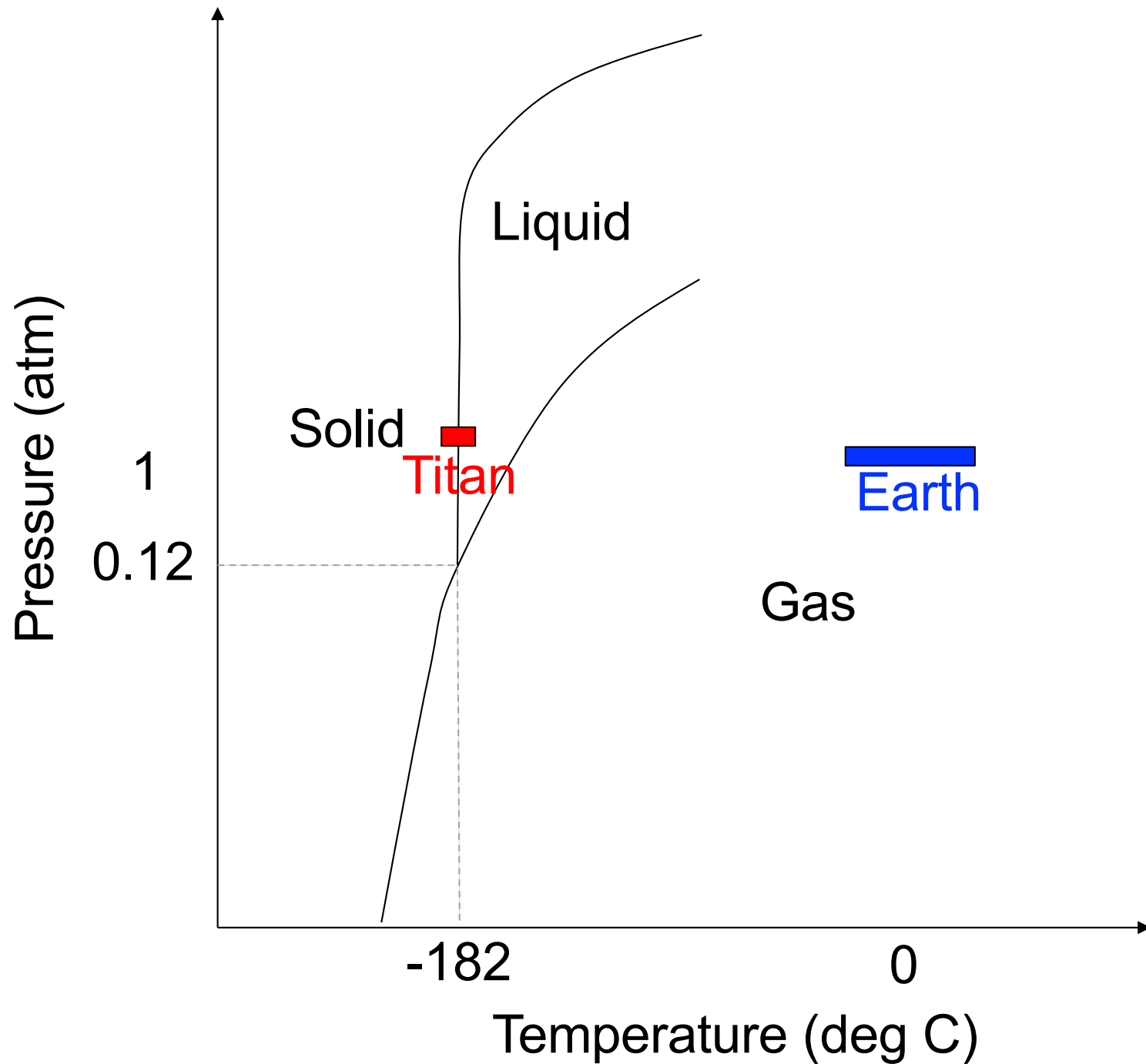


Cloud over Ligeia Mare



Ligeia Mare (the large methane sea)

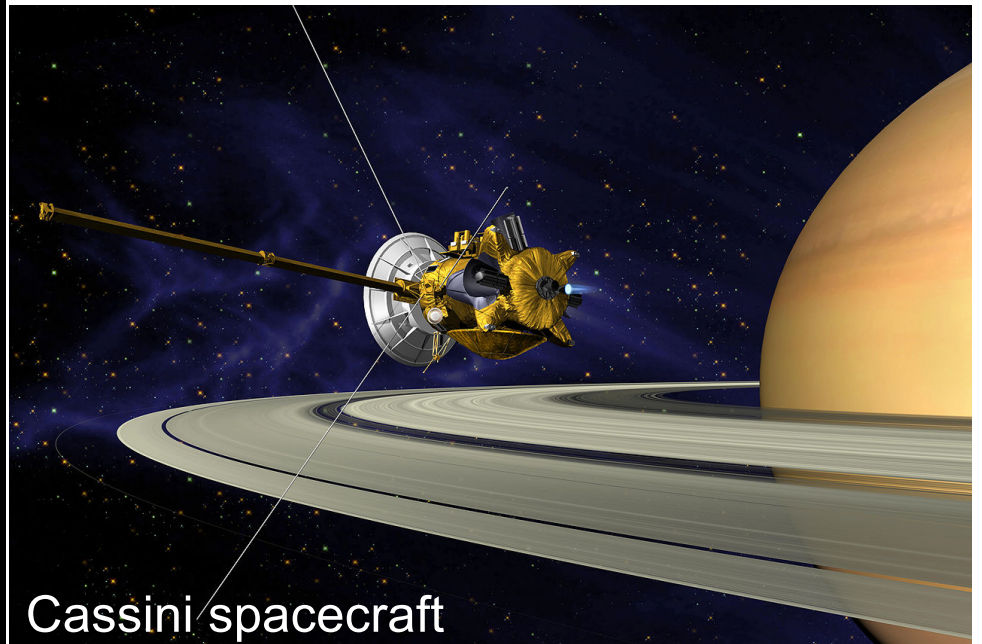
CH₄ phase diagram



Enceladus

- 500 km in diameter
- Mass density 1.6 g/cm^3

Launched in 1997 October 15
Arrived in 2004 June 30
Ended in 2017 September 15



Cassini spacecraft

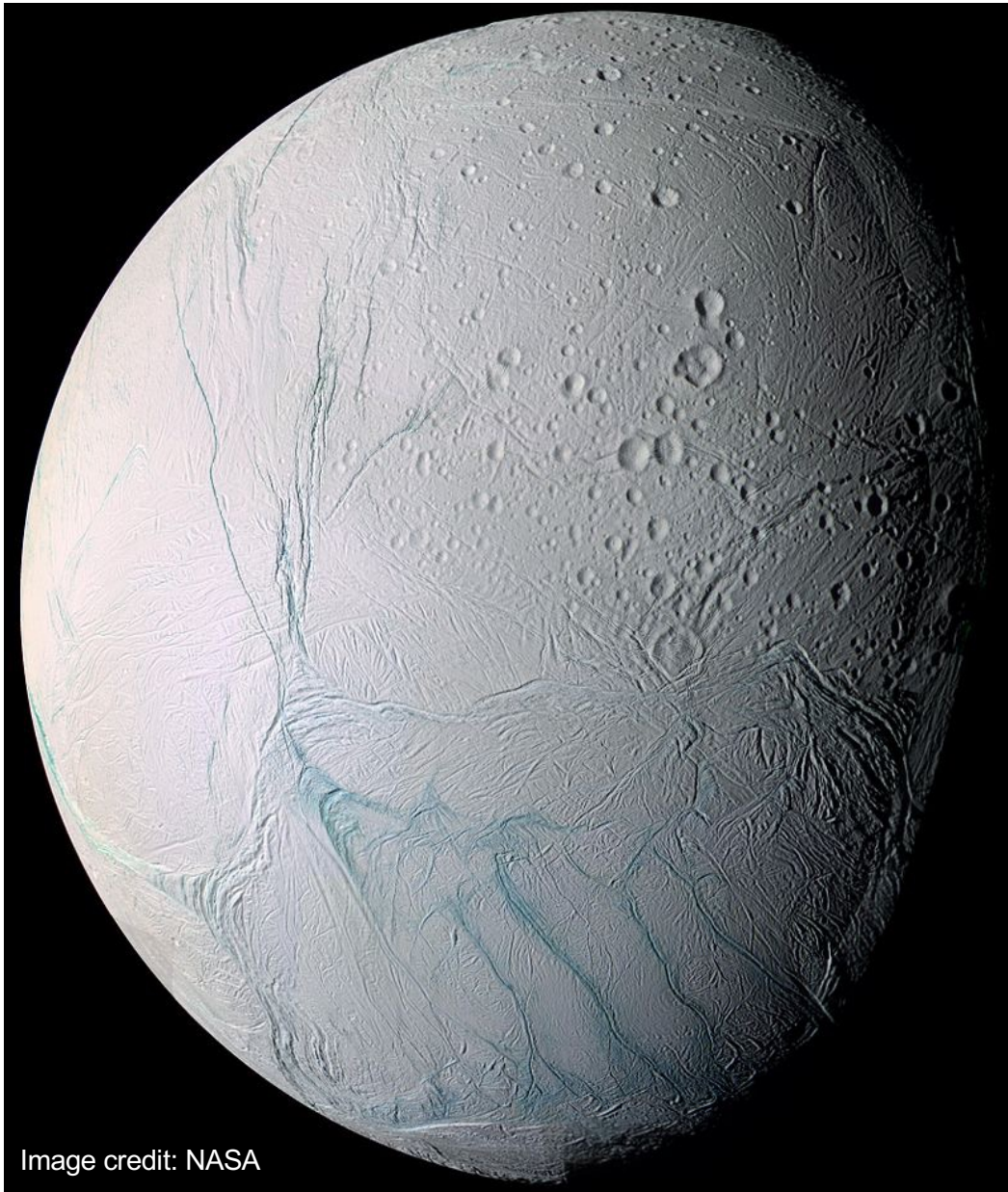
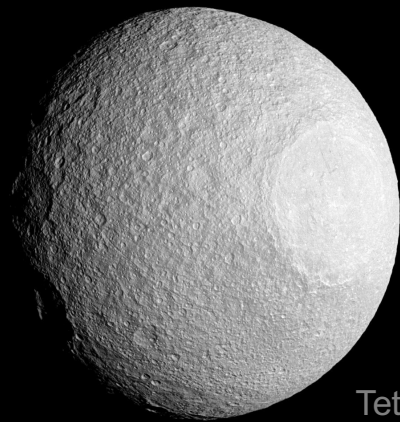


Image credit: NASA

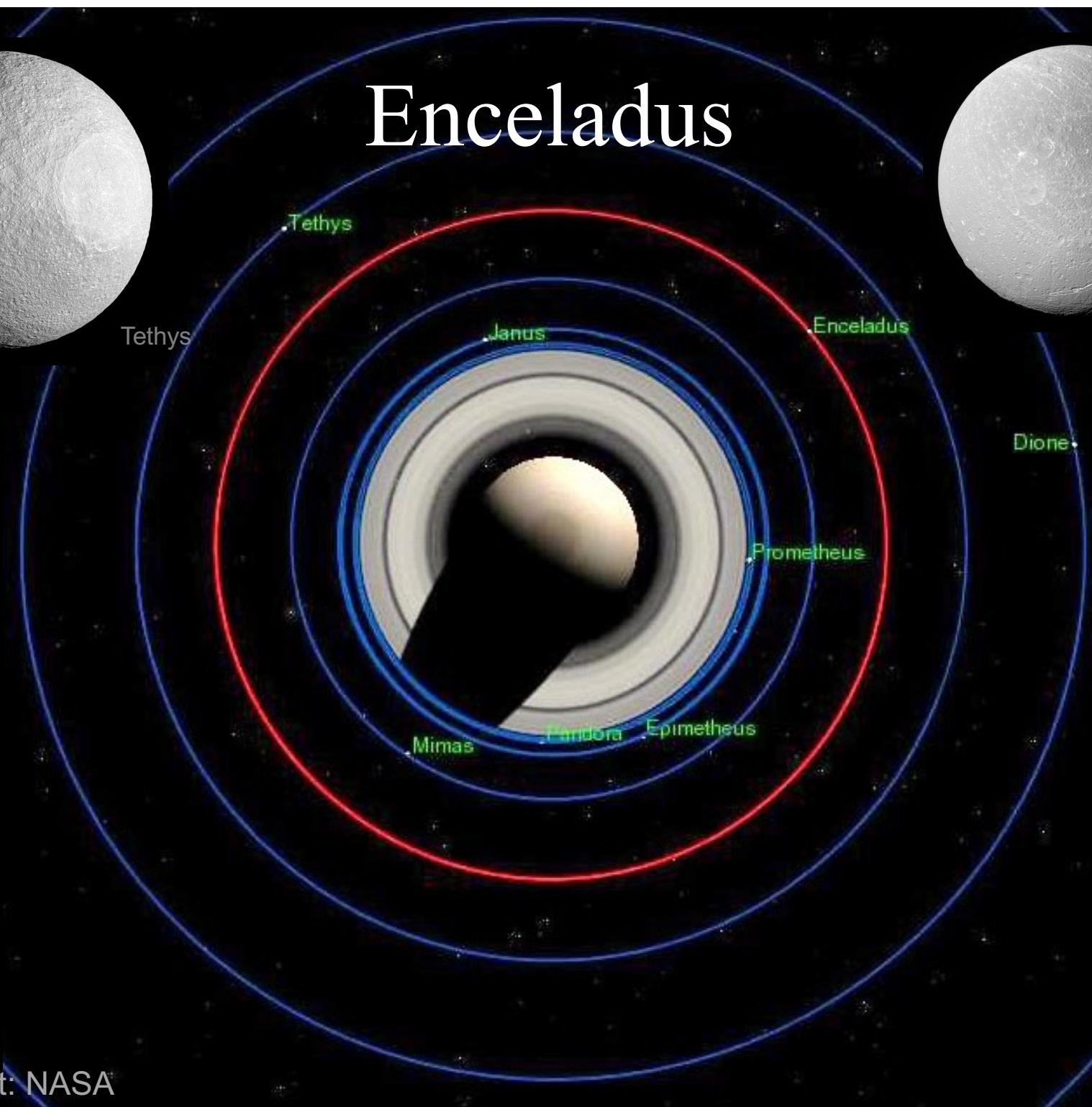
Enceladus



Tethys



Dione



Enceladus

15 km

- H₂O ice & vapor
- NH₃
- CH₄
- Carbonate
- Salt (as a result of interaction between liquid water and rocks)
- Polymeric organic matter

Geyser-like jets

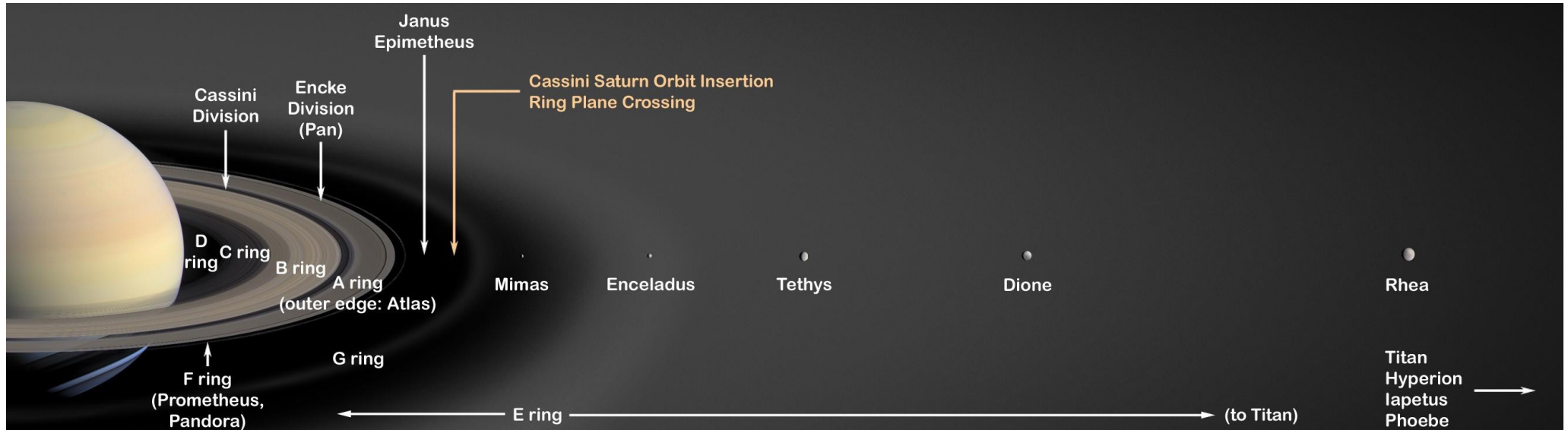
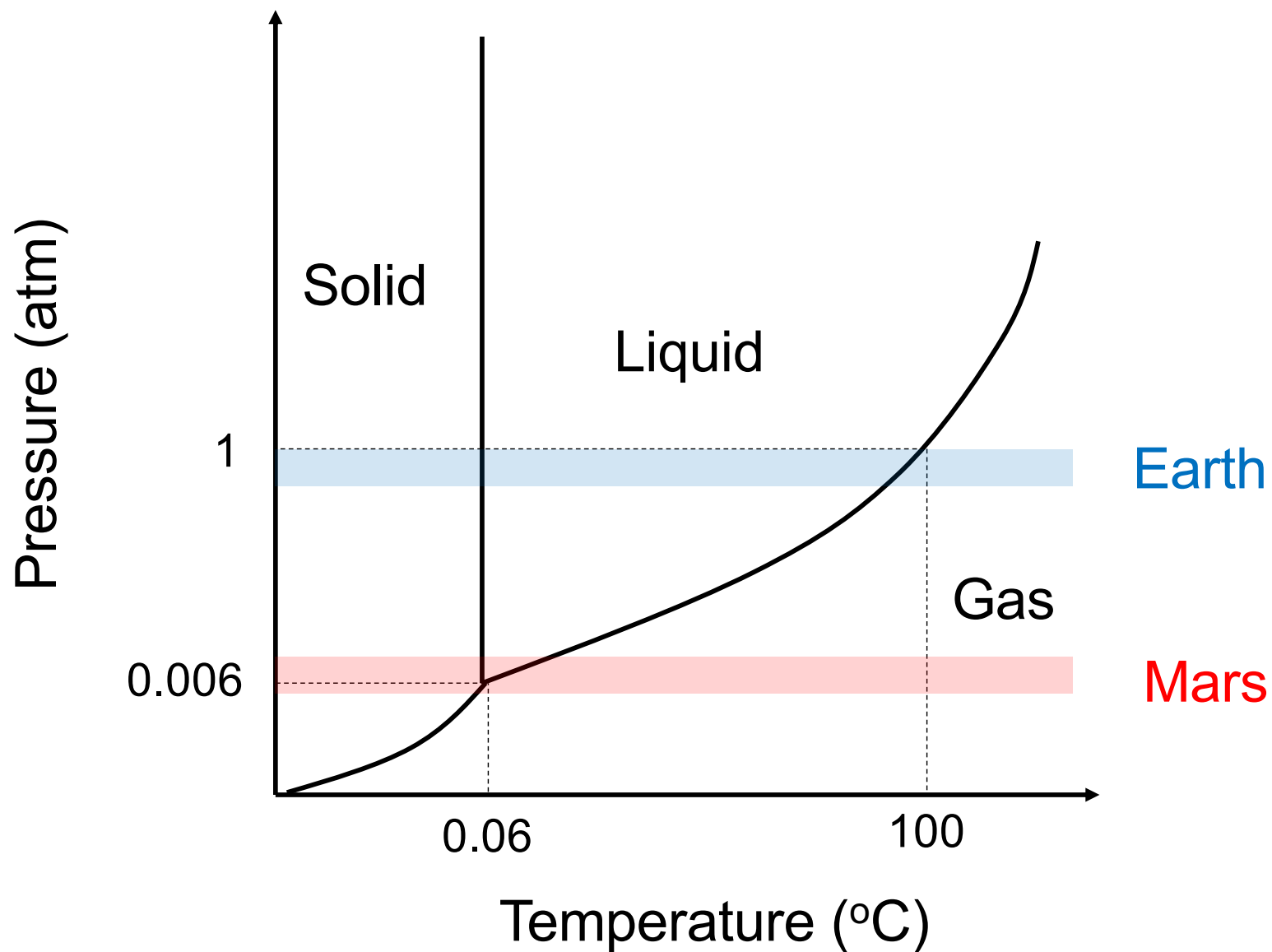
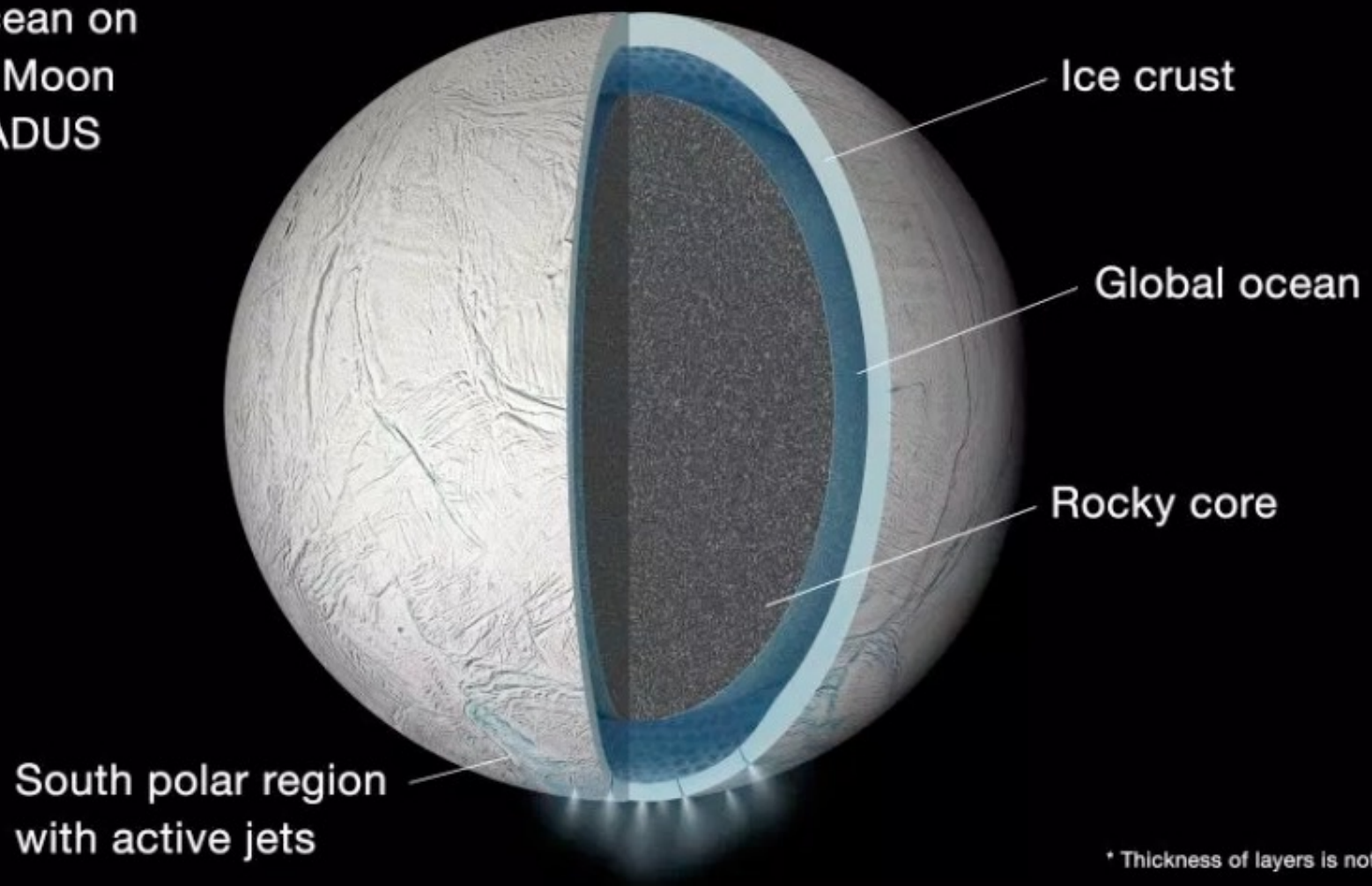


Image credit: NASA

Phase diagram of water



Global Ocean on
Saturn's Moon
ENCELADUS

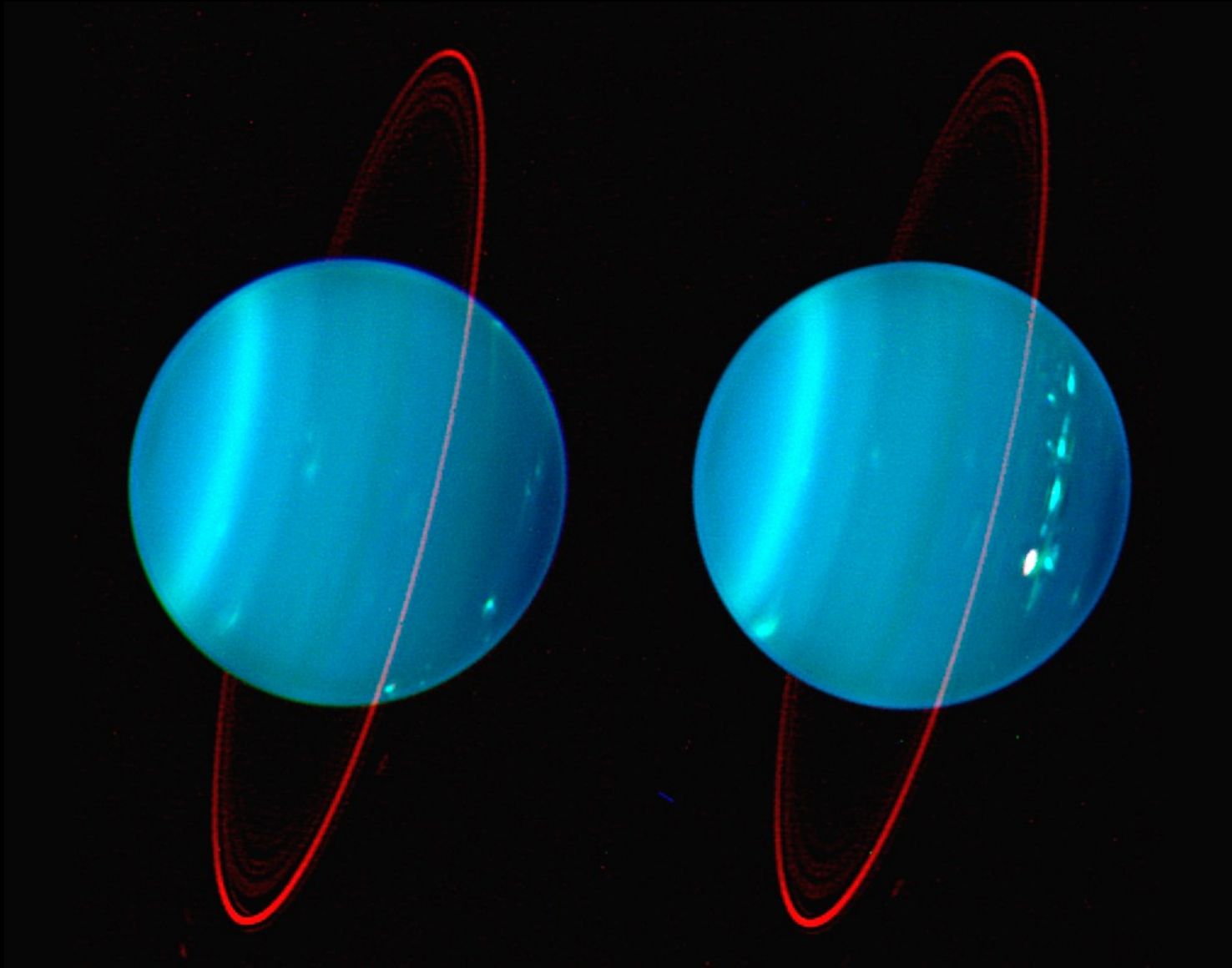


* Thickness of layers is not to scale

Image credit: NASA

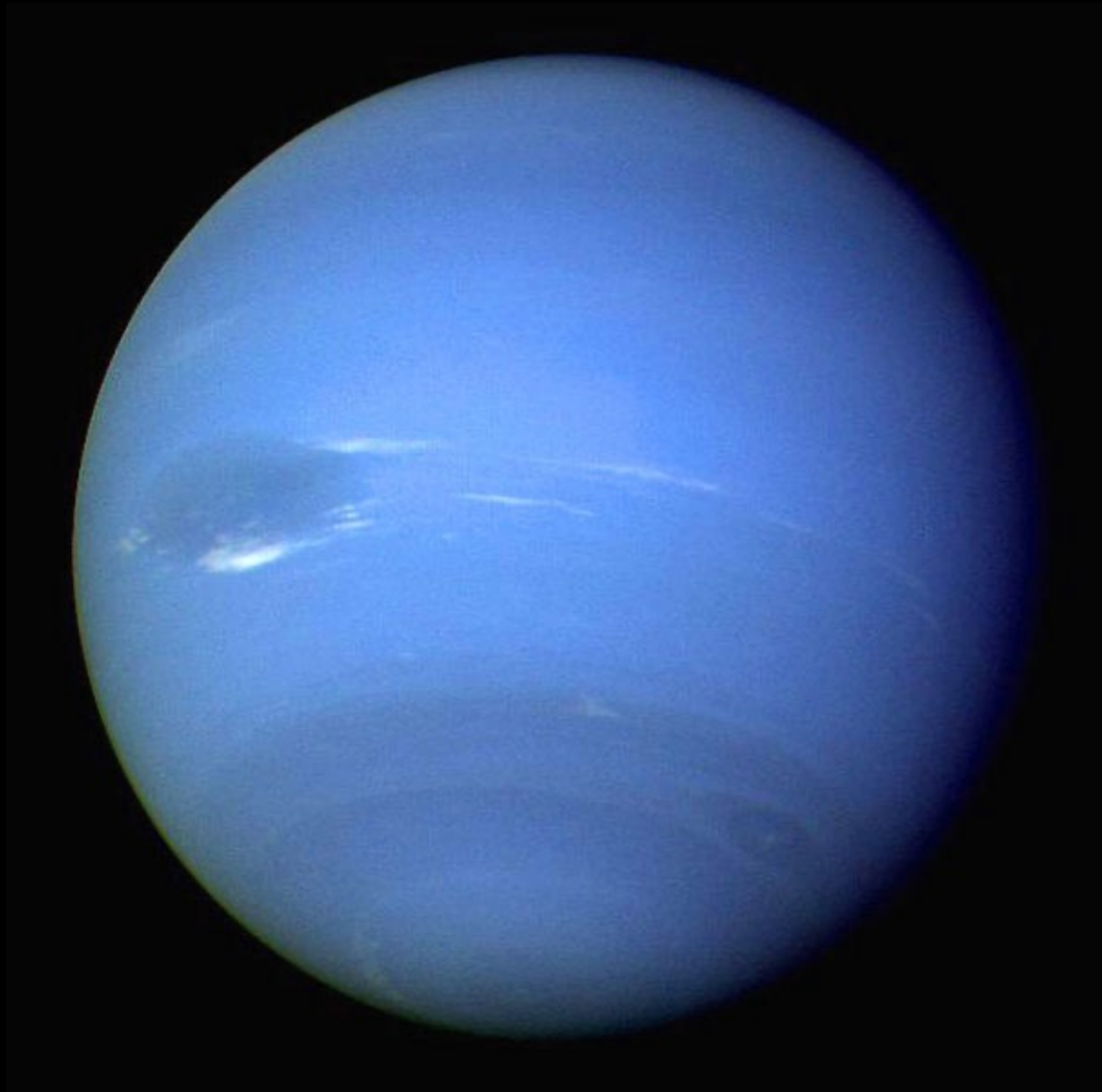
Ice giant planets: Uranus & Neptune

- Dominant composition:
 - Water + Ammonia + Methane ices
 - Only atmosphere contains H, He (in total only minor)
- Uranus:
 - 25% Iron + Silicates
 - 60% Methane + Water + Ammonia
 - 15% Hydrogen + Helium
- Neptune:
 - 20% Iron + Silicates
 - 70% Methane + Water + Ammonia
 - 10% Hydrogen + Helium



Uranus (Near-Infrared Images by Keck Telescope)

Image credit: NASA



Neptune from Voyager 2

Image credit: NASA



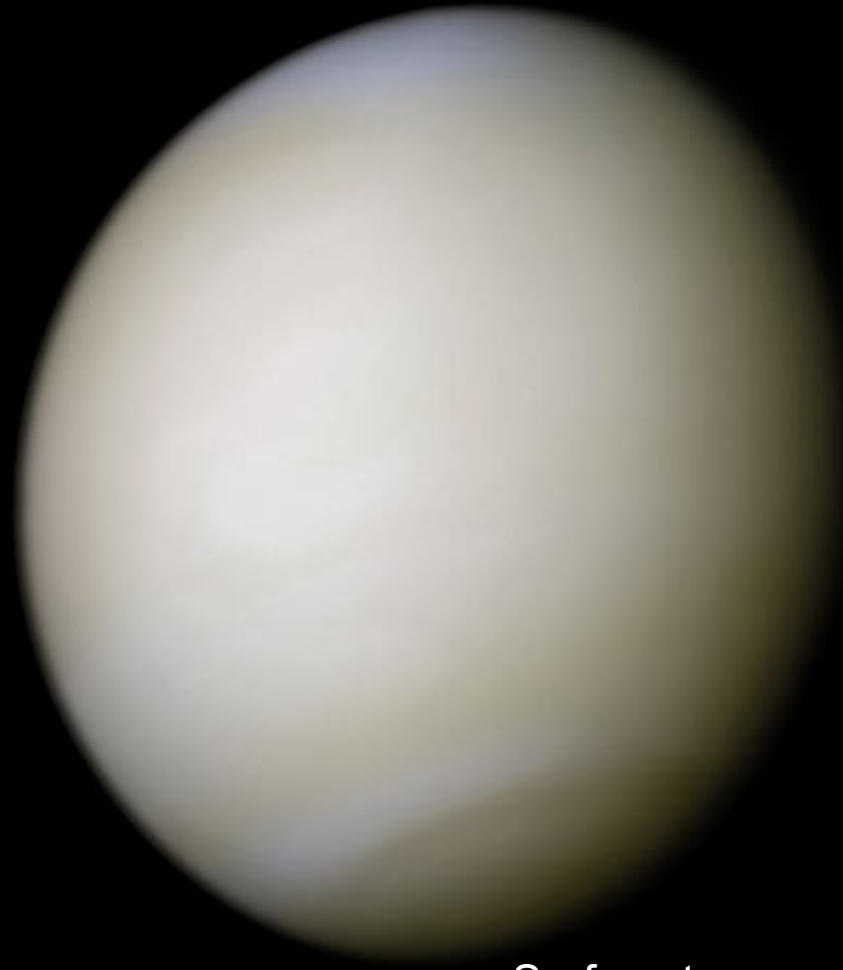
Image credit: NASA

Surface temperature on equator
So many craters like moon

700K (day), 100K (night)

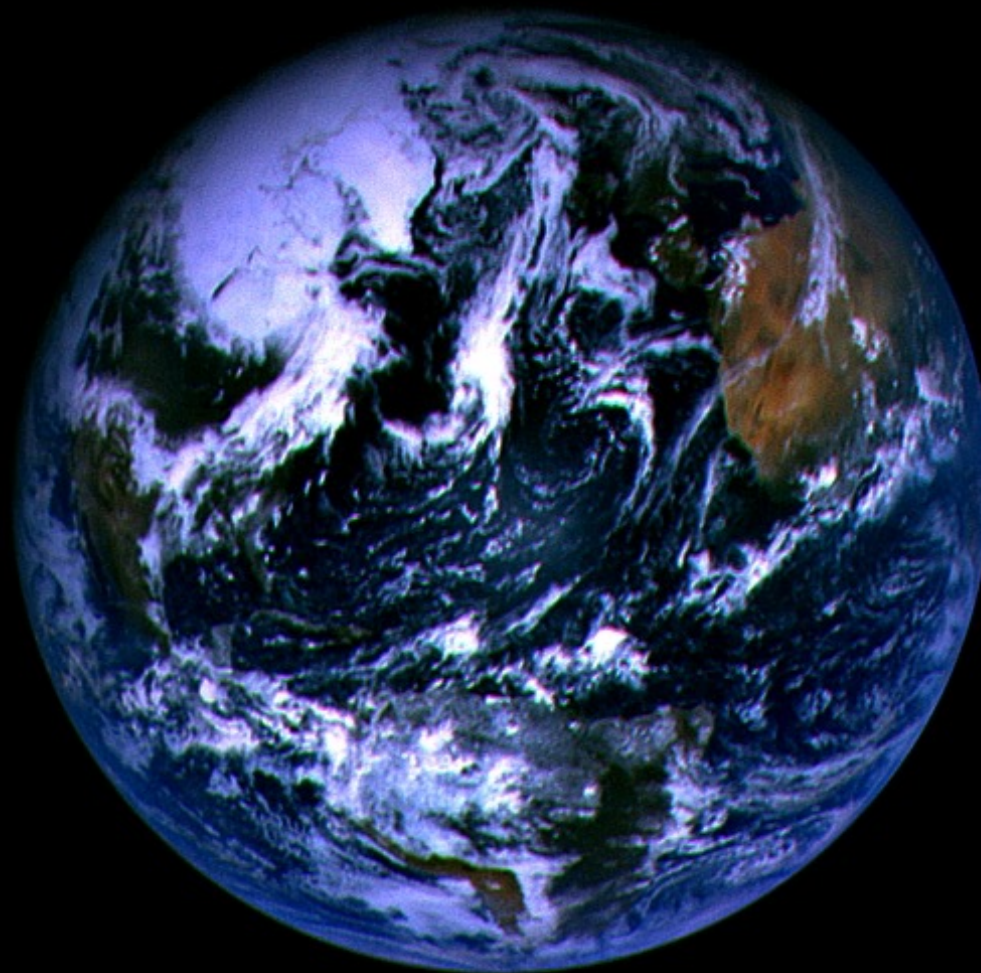


Due to the thick cloud, only 20% of solar flux arrives on the surface.



Surface temperature	735K (average)
	770K (max)
Surface pressure	9.3 Mpa (90 atm)
Atmospheric composition	CO ₂ 96.5%
	N ₂ 3.5%
	SO ₂ 0.015%

Image credit: NASA



Surface temperature	180-330K (average 287K)	
Surface pressure	100 kPa	
Atmospheric composition	N ₂	78%
	O ₂	21%
	H ₂ O	~1%



Surface temperature	180-270 K
Surface pressure	0.4-0.9 kPa
Atmospheric composition	CO ₂ 95.3%
	N ₂ 2.7 %

Image credit: NASA

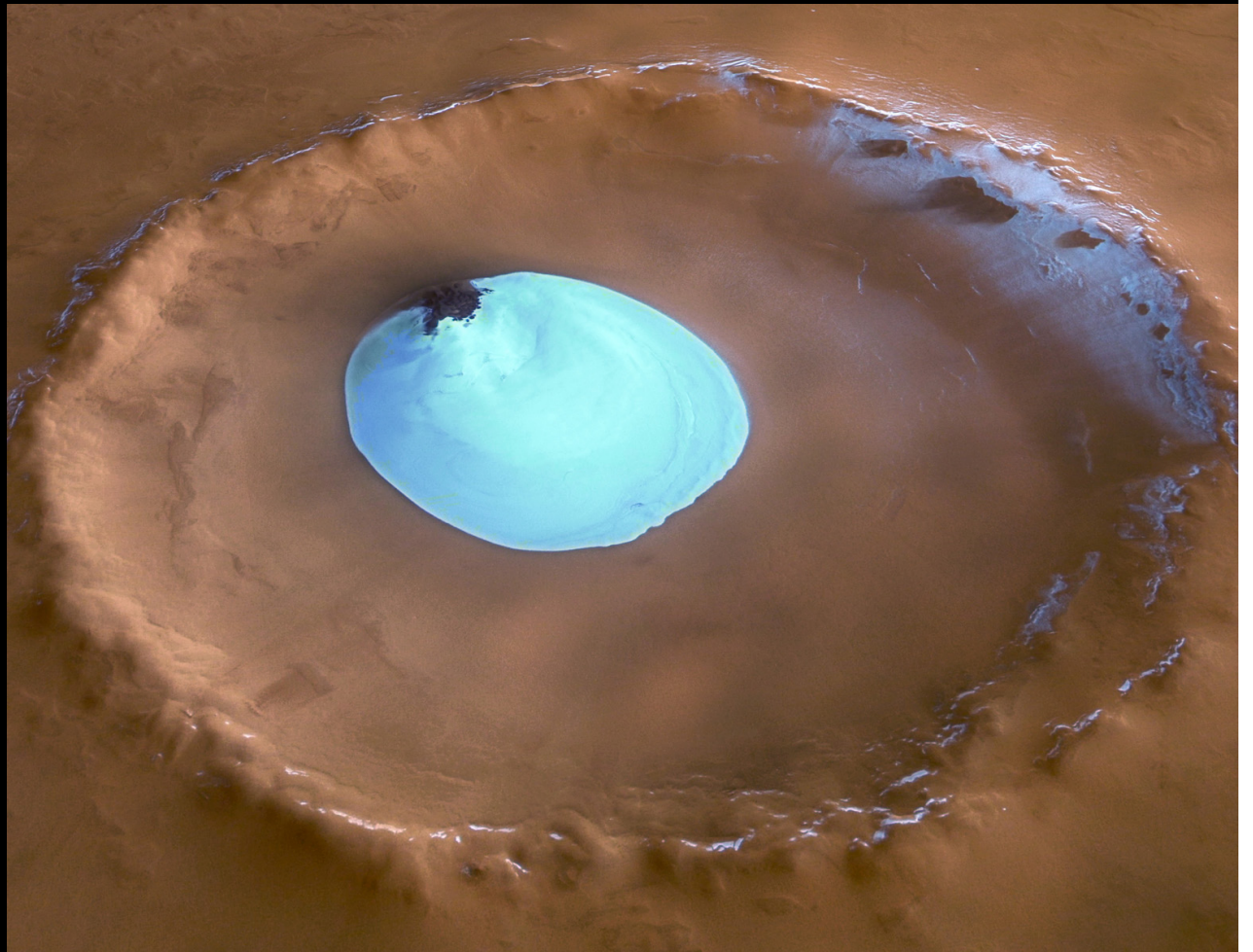
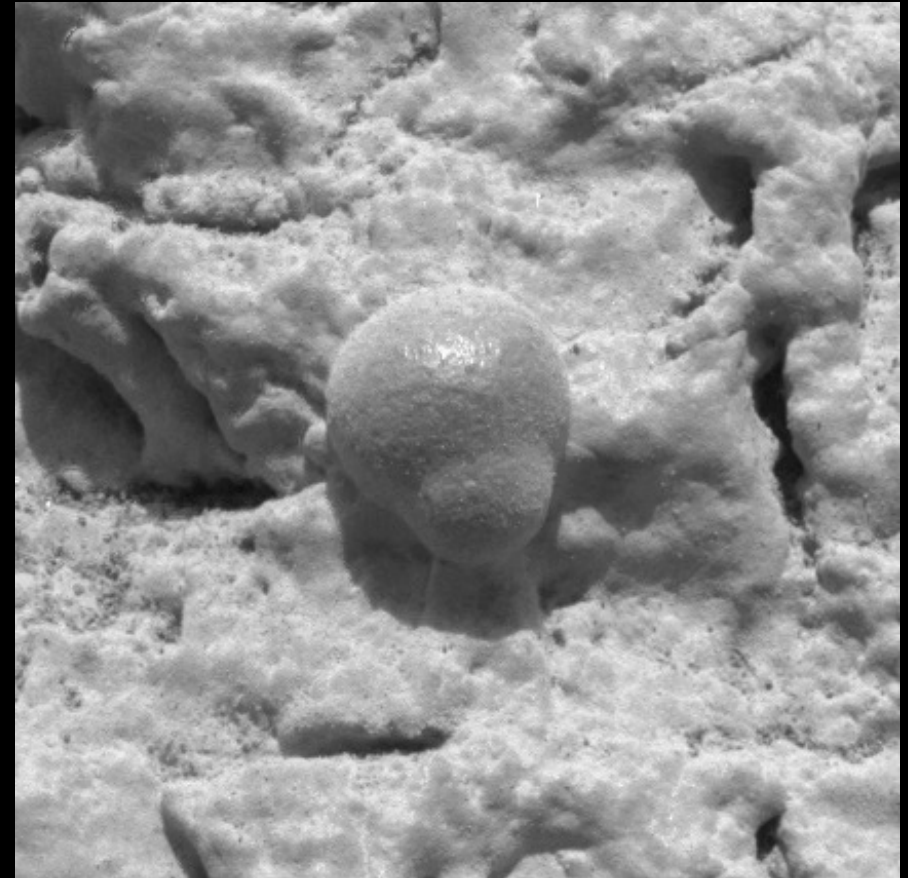
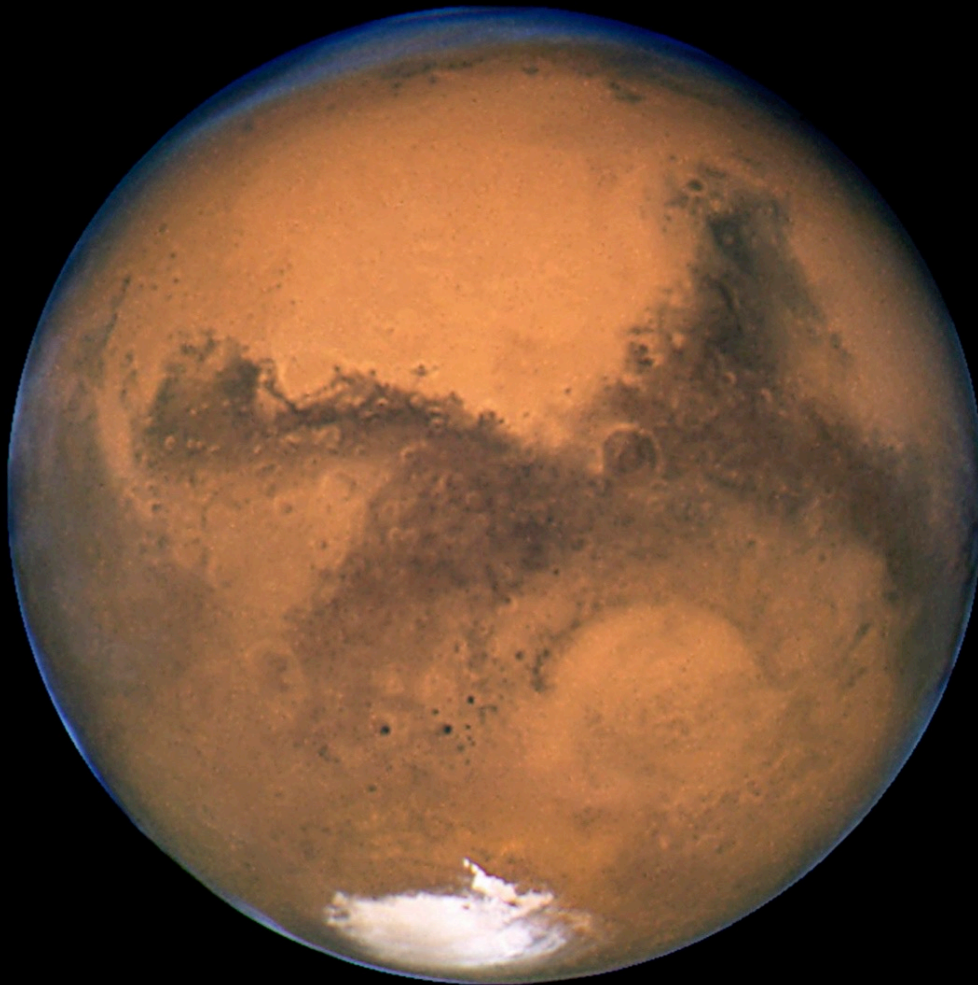


Image credit: ESA's Mars Express

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Planet Mars (indicative past liquid water?)

Outflow Channels

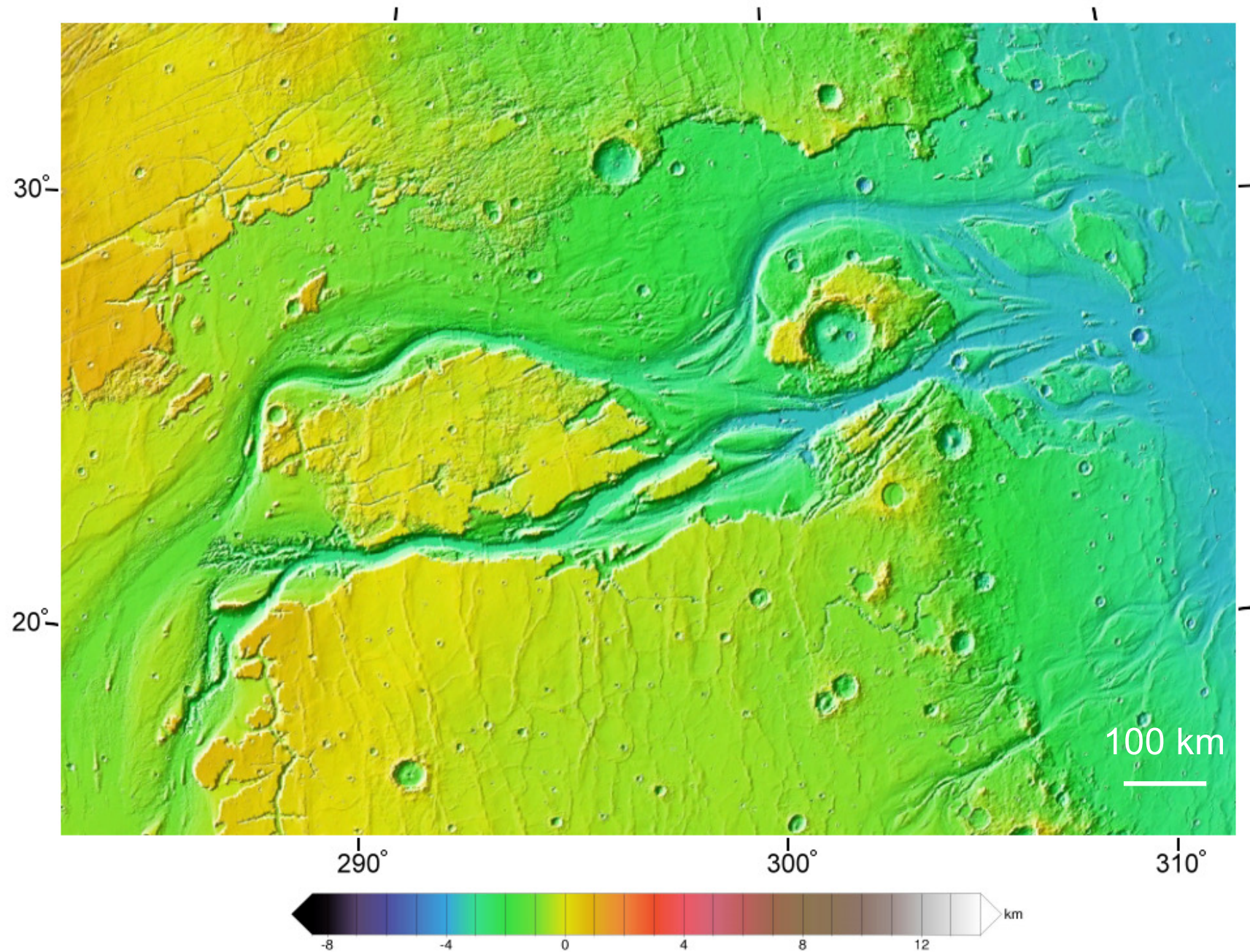
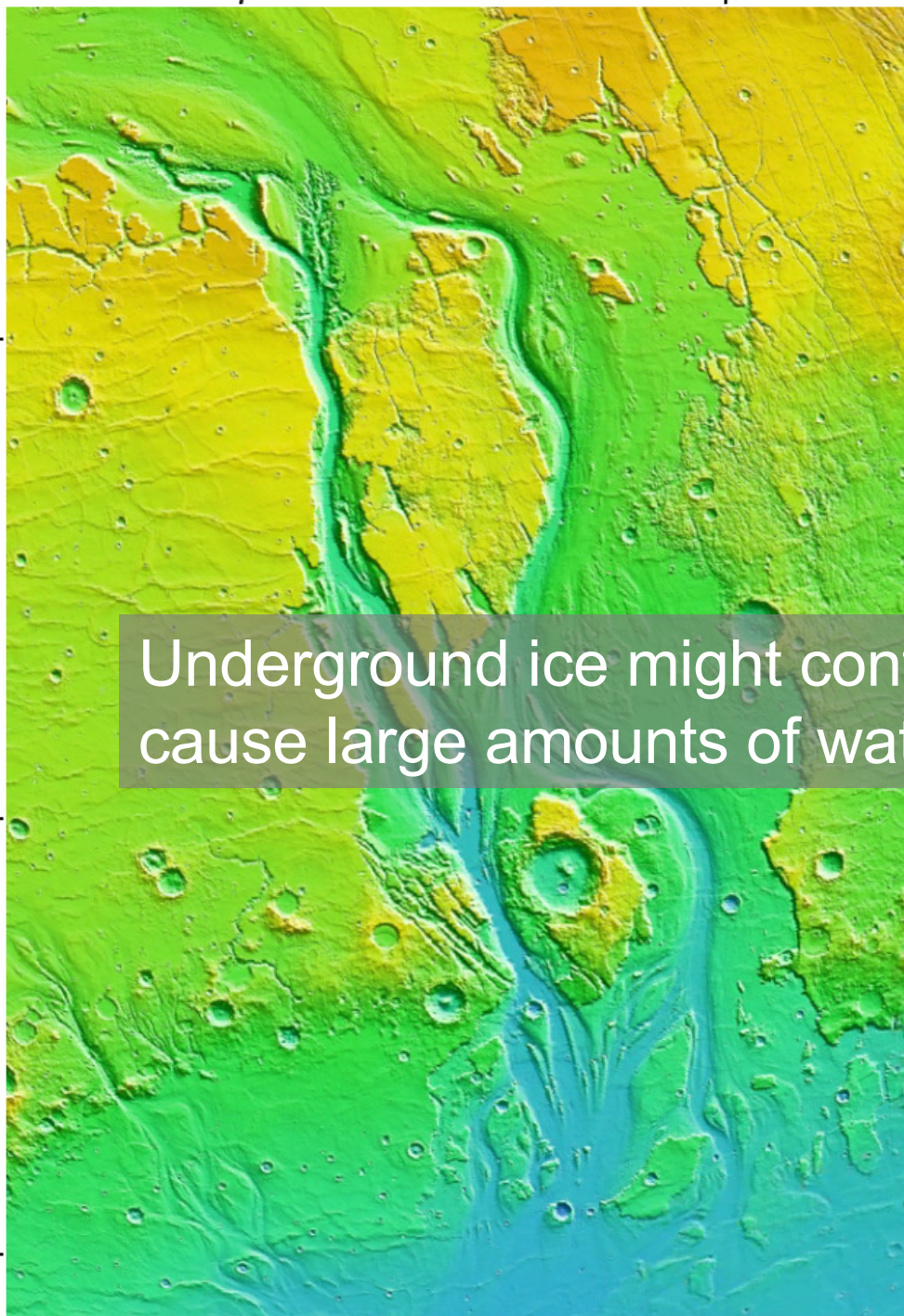
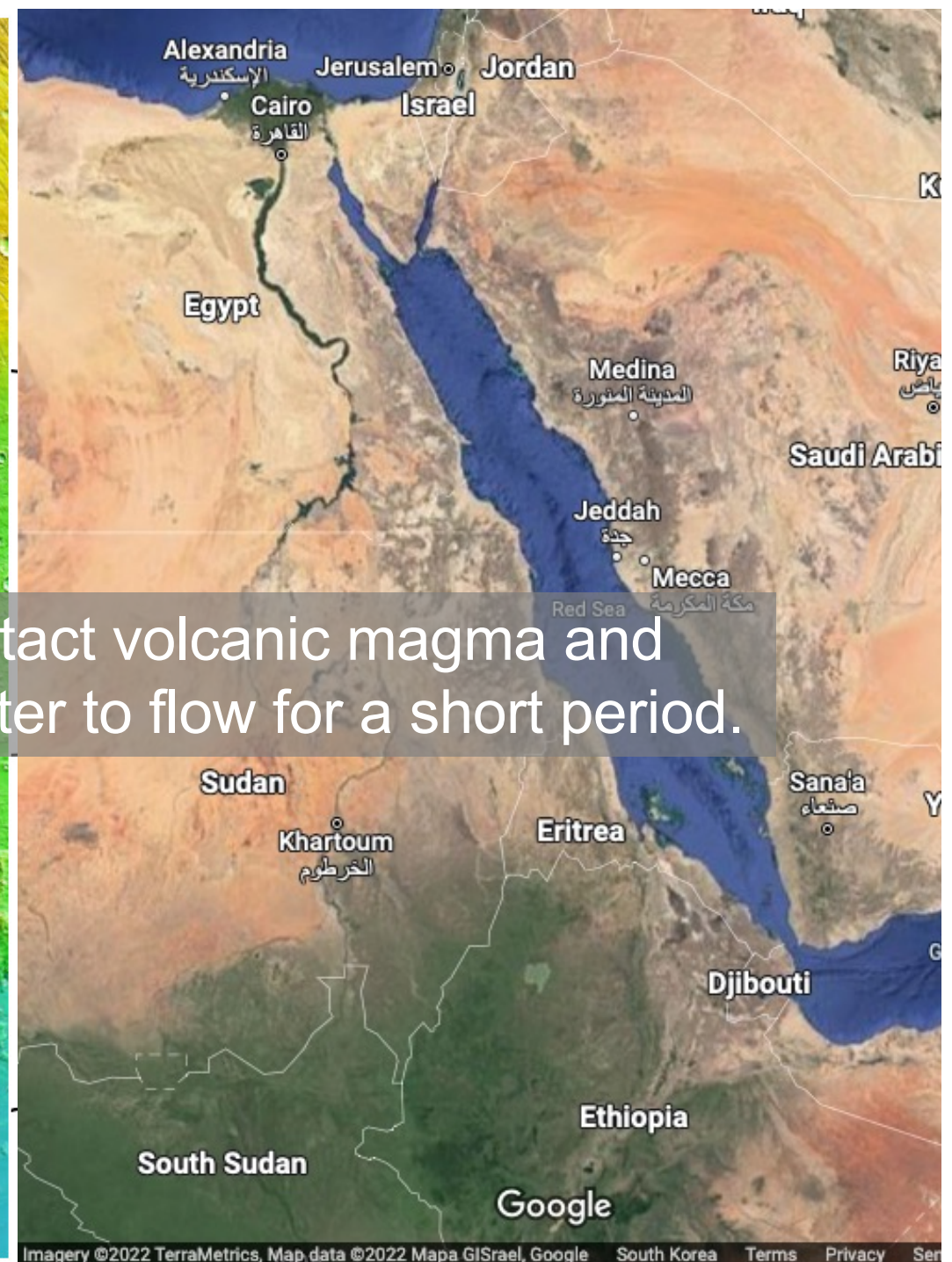


Image credit: NASA

a signiture of land erosion by flowing water

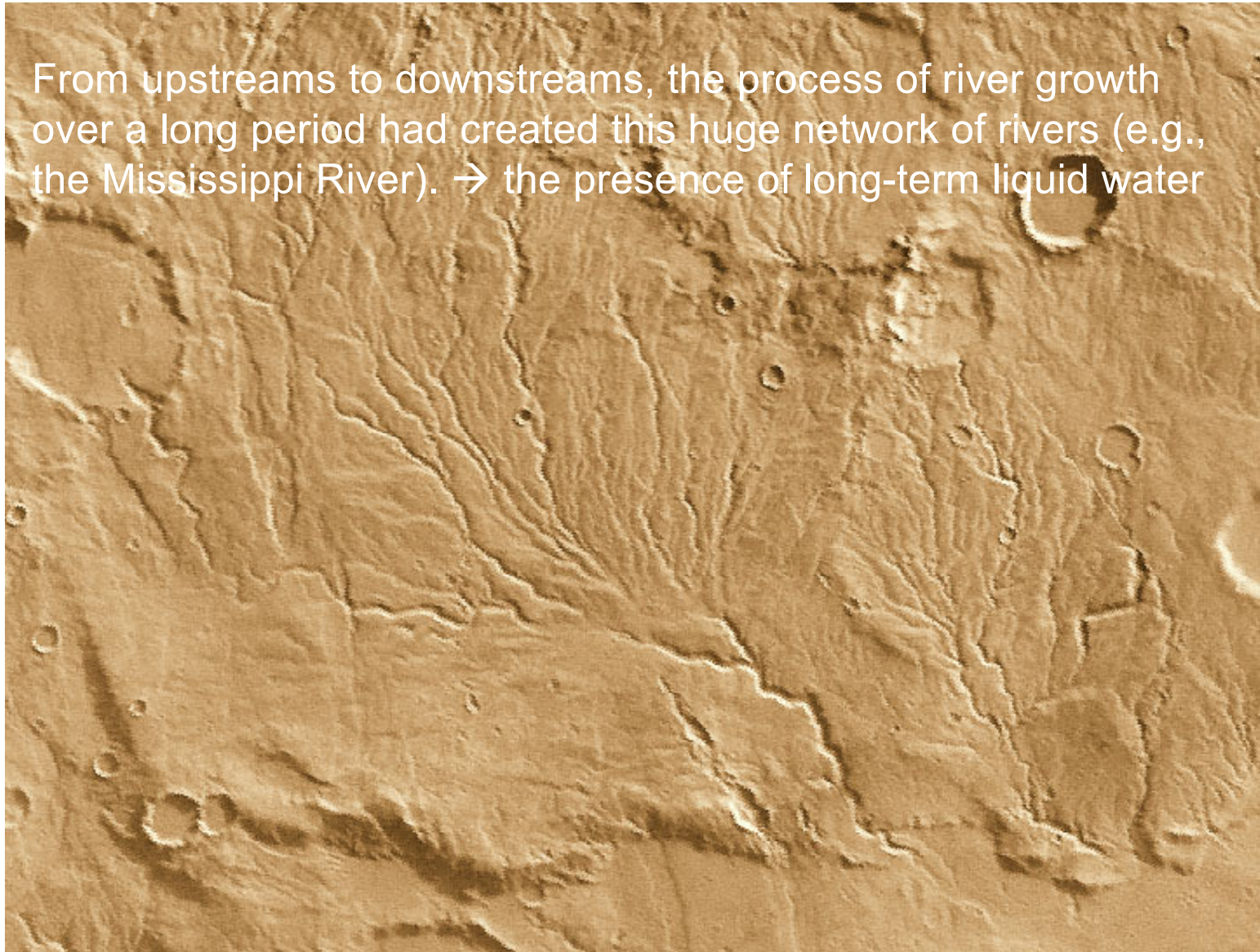


Underground ice might contact volcanic magma and cause large amounts of water to flow for a short period.



Valley Network

From upstreams to downstreams, the process of river growth over a long period had created this huge network of rivers (e.g., the Mississippi River). → the presence of long-term liquid water



H₂O phase diagram

